

Electromagnetism

1 Introduction

1.1 Charges and currents

Electric charge is a physical property of elementary particles. It is

- (i) positive, negative or zero;
- (ii) quantised, being an integer multiple of the elementary charge e ;
- (iii) conserved, even if particles are created or destroyed.

By convention, the electron has charge $-e$. The proton has charge $+e$ and the neutron has zero charge. On macroscopic scales, the number of particles is so large that charge can be considered to have a continuous electric charge density $\rho(\mathbf{x}, t)$. The total charge in a volume V is

$$Q = \int_V \rho dV.$$

The electric current density $\mathbf{J}(\mathbf{x}, t)$ is the flux of electric charge per unit area. The current flowing through a surface S is

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}.$$

Consider a time-independent volume V with boundary S . Since charge is conserved,

$$\begin{aligned} \frac{dQ}{dt} &= -I \\ \frac{d}{dt} \int_V \rho dV + \int_S \mathbf{J} \cdot d\mathbf{S} &= 0 \\ \int_V \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} \right) dV &= 0. \end{aligned}$$

Since this is true for any V ,

1.1 – Charge conservation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

This equation of charge conservation has the typical form of a conservation law.

The discrete charge distribution of a single particle of charge q_i and position vector $\mathbf{x}_i(t)$ is

$$\rho(\mathbf{x}, t) = q_i \delta(\mathbf{x} - \mathbf{x}_i(t)) \text{ with } \mathbf{J}(\mathbf{x}, t) = q_i \dot{\mathbf{x}}_i \delta(\mathbf{x} - \mathbf{x}_i(t)).$$

For N particles, it is

$$\rho(\mathbf{x}, t) = \sum_{i=1}^N q_i \delta(\mathbf{x} - \mathbf{x}_i(t)) \text{ with } \mathbf{J}(\mathbf{x}, t) = \sum_{i=1}^N q_i \dot{\mathbf{x}}_i \delta(\mathbf{x} - \mathbf{x}_i(t)).$$

We can verify that these satisfy 1.1.

1.2 Fields and forces

Electromagnetism is a field theory. Charge particles interact not directly but by generating fields around them that are experienced by other charged particles. In general, we have two time-dependent vector fields, the electric field $\mathbf{E}(\mathbf{x}, t)$ and the magnetic field $\mathbf{B}(\mathbf{x}, t)$.

1.2 – Lorentz force

The Lorentz force on a particle of charge q and velocity $\mathbf{v}$ is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

1.3 Maxwell's equations

In this course, we will explore some consequences of

1.3 – Maxwell's equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

We list some properties of Maxwell's equations.

- (i) They are coupled linear partial differential equations in space and time.
- (ii) They involve two positive constants, the vacuum permittivity ϵ_0 and the vacuum permeability μ_0 .
- (iii) Charges ρ and currents $\mathbf{J}$ are the sources of electromagnetic fields.
- (iv) Each equation has an equivalent integral form related via the divergence theorem or Stokes' theorem. These will appear later in the course.
- (v) These are the vacuum equations that apply on microscopic scales or in a vacuum. A related macroscopic version applies in media, which is discussed in Part II Electrodynamics.
- (vi) The equations are consistent with each other and with charge conservation. $\nabla \cdot (\text{M3})$ agrees with $\partial/\partial t (\text{M2})$. Also,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = \frac{\partial}{\partial t} (\epsilon_0 \nabla \cdot \mathbf{E}) + \nabla \cdot \left(-\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \frac{1}{\mu_0} \nabla \times \mathbf{B} \right) = 0.$$

1.4 Units

The SI unit of electric charge is the coulomb C. The elementary charge is exactly $e = 1.602176634 \times 10^{-19}$ C. The SI unit of electric current is the ampere or amp A, equal to 1 C s^{-1} . The SI base units needed in electromagnetism are the second s, metre m, kilogram kg and ampere A. From the Lorentz force law, we see that the units of $\mathbf{E}$ and $\mathbf{B}$ must be $\text{kg m s}^{-3} \text{ A}^{-1}$ and $\text{kg s}^{-2} \text{ A}^{-1}$ respectively. The unit of $\mathbf{B}$ is also called the tesla T. From Maxwell's equations, we can work out the units of ϵ_0 and μ_0 . The experimentally determined values are

$\epsilon_0 = 8.854 \times 10^{-12} \text{ kg}^{-1} \text{ m}^{-3} \text{ s}^4 \text{ A}^2$ and $\mu_0 = 1.256 \times 10^{-6} \text{ kg m s}^{-2} \text{ A}^{-2}$. The speed of light is exactly

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 299792458 \text{ m s}^{-1}.$$

2 Electrostatics

In a time-independent situation, Maxwell's equations reduce to

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho / \epsilon_0 & \nabla \times \mathbf{E} &= \mathbf{0} \\ \nabla \cdot \mathbf{B} &= 0 & \nabla \times \mathbf{B} &= \mu_0 \mathbf{J} \end{aligned}$$

Since $\mathbf{E}$ and $\mathbf{B}$ are decoupled, we can study them separately. Electrostatics is the study of the electric field generated by a stationary charge distribution.

2.1 Gauss's law

Consider a closed surface S enclosing a volume V . Integrate (M1) over V and use the divergence theorem to obtain

2.1 – Gauss's law

$$\int_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0}$$

where $Q = \int_V \rho dV$ is the total charge in V .

Gauss's law is the integral version of (M1) and is valid generally. We can interpret this as electric flux $\propto$ total charge enclosed. In special situations, we can use Gauss's law together with symmetry to deduce $\mathbf{E}$ from ρ by choosing the Gaussian surface S appropriately.

2.1.1 Spherical symmetry

Consider a spherically symmetric charge distribution $\rho(r)$ in spherical polar coordinates, with total charge Q contained within an outer radius R . To have spherical symmetry, the electric field should have the form $\mathbf{E} = E(r)\mathbf{e}_r$. This will satisfy (M3') as required. To find $E(r)$, we apply Gauss's law to a sphere of radius r .

If $r > R$, then

$$\begin{aligned} \int_S \mathbf{E} \cdot d\mathbf{S} &= E(r) \int_S \mathbf{e}_r \cdot d\mathbf{S} = E(r) \int_S dS = 4\pi r^2 E(r) = \frac{Q}{\epsilon_0} \\ \mathbf{E} &= \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{e}_r. \end{aligned}$$

Thus, the external electric field of a spherically symmetric body depends only on the total charge.

2.2 – Coulomb force

The Lorentz force on a particle of charge q in $r > R$ is

$$\mathbf{F} = q\mathbf{E} = \frac{Qq}{4\pi\epsilon_0 r^2} \mathbf{e}_r.$$

The force is repulsive if the charges have the same sign and attractive if they have opposite signs. In the limit $R \rightarrow 0$, we obtain the electric field of a point charge Q , corresponding to $\rho = Q\delta(\mathbf{x})$.

There is a close analogy between the Coulomb force and the gravitational force between massive particles, $\mathbf{F} = -GMm/r^2 \mathbf{e}_r$. Both involve an inverse-square law and the product of the charges or masses. However,

- (i) while gravity is always attractive, electric forces can be repulsive or attractive.
- (ii) gravity is very much weaker. For two protons, the ratio of the electric force to gravitational force is $e^2/(4\pi\epsilon_0 Gm_p^2) \approx 10^{36}$. On the atomic scale, gravity is irrelevant. But the positive and negative charges balance so accurately that on the planetary scale, gravity is dominant.

2.1.2 Cylindrical symmetry

Consider a cylindrically symmetric charge distribution $\rho(r)$ in cylindrical polar coordinates, with total charge λ per unit length contained within an outer radius R . To have cylindrical symmetry, $\mathbf{E} = E(r)\mathbf{e}_r$ which satisfies (M3') as required. To find $E(r)$, we apply Gauss's law to a cylinder of radius r and arbitrary length L .

If $r > R$, then

$$\int_S \mathbf{E} \cdot d\mathbf{S} = E(r) \int_S \mathbf{e}_r \cdot d\mathbf{S} = E(r) \int_{S_{\text{curved}}} dS = 2\pi r L E(r) = \frac{\lambda L}{\epsilon_0}$$

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 r} \mathbf{e}_r.$$

In the limit $R \rightarrow 0$, we obtain the electric field of a line charge λ per unit length, corresponding to $\rho = \lambda\delta(x)\delta(y)$.

2.1.3 Planar symmetry

Consider a planar charge distribution $\rho(z)$ in Cartesian coordinates, with total charge σ per unit area contained within a region $-d < z < d$ of thickness $2d$. Assume that there is reflectional symmetry so $\rho(z)$ is even. To have planar symmetry, $\mathbf{E} = E(z)\mathbf{e}_z$ which satisfies (M3'). Reflectional symmetry implies $E(-z) = -E(z)$. To find $E(z)$ for $z > 0$, we apply Gauss's law to a Gaussian pillbox of height $2z$ and arbitrary area A .

If $z > d$, then

$$\int_S \mathbf{E} \cdot d\mathbf{S} = E(z)A - E(-z)A = 2AE(z) = \frac{\sigma A}{\epsilon_0}$$

$$\mathbf{E} = \begin{cases} (\sigma/2\epsilon_0)\mathbf{e}_z & \text{if } z > d \\ -(\sigma/2\epsilon_0)\mathbf{e}_z & \text{if } z < -d. \end{cases}$$

In the limit $d \rightarrow 0$, we obtain the electric field of a surface charge σ per unit area, corresponding to $\rho = \sigma\delta(z)$.

2.1.4 Surface charge and discontinuity

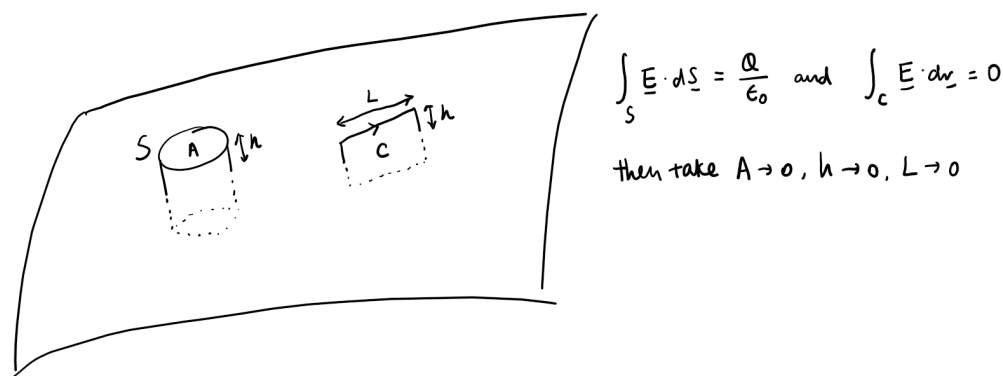
Let $\mathbf{n}$ be a unit vector normal to the charged surface, pointing from region I to region II. In our example, $\mathbf{n} = \mathbf{e}_z$. The discontinuity in $\mathbf{E}$ is given by

$$[\mathbf{n} \cdot \mathbf{E}] = \frac{\sigma}{\epsilon_0}$$

where σ is the surface charge density and $[X] = X_2 - X_1$ denotes a discontinuity. The tangential components are continuous

$$[\mathbf{n} \times \mathbf{E}] = \mathbf{0}$$

by Stokes' theorem. These equations apply to any surface charge, even if the surface is curved and the charge is non-uniform.



2.2 The electrostatic potential

For general $\rho(\mathbf{x})$, we cannot determine $\mathbf{E}(\mathbf{x})$ using Gauss's law alone. (M3') implies that $\mathbf{E}$ can be written in terms of an

2.3 – Electrostatic potential $\Phi(\mathbf{x})$

$$\mathbf{E} = -\nabla\Phi$$

The potential difference or voltage between two points $\mathbf{x}_1$ and $\mathbf{x}_2$ is

$$\Phi(\mathbf{x}_2) - \Phi(\mathbf{x}_1) = \int d\Phi = - \int_{\mathbf{x}_1}^{\mathbf{x}_2} \mathbf{E} \cdot d\mathbf{x}$$

and is path-independent because $\nabla \times \mathbf{E} = \mathbf{0}$. The electric force on a particle of charge q is $\mathbf{F} = q\mathbf{E} = -q\nabla\Phi$. This is a conservative force associated with the

2.4 – Potential energy

$$U(\mathbf{x}) = q\Phi(\mathbf{x})$$

(M1) implies that Φ satisfies Poisson's equation

$$-\nabla^2\Phi = \frac{\rho}{\epsilon_0}.$$

The solution can be written as an integral (over all space and assuming decay at infinity).

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_{\mathbb{R}^3} \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3\mathbf{x}'.$$

This is the convolution of $\rho(\mathbf{x})$ with the potential of a unit point charge $1/(4\pi\epsilon_0|\mathbf{x}|)$, which is the solution of $-\nabla^2\Phi = \delta(\mathbf{x})/\epsilon_0$ satisfying $\Phi \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$. Note that $\mathbf{E}$ is unaffected if we add an arbitrary constant to Φ . We usually choose this such that $\Phi \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$. If $\rho(\mathbf{x})$ does not decay sufficiently rapidly, this may not be possible. For example, a line charge has $E_r \propto 1/r$ and $\Phi \propto \ln r$.

2.2.1 Point charge

The potential due to a point charge q at the origin is

$$\Phi(\mathbf{x}) = \frac{q}{4\pi\epsilon_0|\mathbf{x}|} = \frac{q}{4\pi\epsilon_0 r}.$$

2.2.2 Electric dipole

Consider two equal and opposite charges at different positions. Without loss of generality, we can consider a charge of $-q$ at $\mathbf{x} = \mathbf{0}$ and a charge of $+q$ at $\mathbf{x} = \mathbf{d}$. The potential due to the dipole is

$$\Phi(\mathbf{x}) = \frac{q}{4\pi\epsilon_0} \left(-\frac{1}{|\mathbf{x}|} + \frac{1}{|\mathbf{x} - \mathbf{d}|} \right).$$

We apply Taylor's theorem for a scalar field

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + (\mathbf{h} \cdot \nabla)f(\mathbf{x}) + \frac{1}{2}(\mathbf{h} \cdot \nabla)^2 f(\mathbf{x}) + \mathcal{O}(|\mathbf{h}|^3)$$

to $f(\mathbf{x}) = 1/|\mathbf{x}| = 1/r$.

$$\begin{aligned} \Phi(\mathbf{x}) &= \frac{q}{4\pi\epsilon_0} \left(-\frac{1}{r} + \frac{1}{r} - (\mathbf{d} \cdot \nabla) \frac{1}{r} + \mathcal{O}(|\mathbf{d}|^2) \right) \\ &= \frac{q}{4\pi\epsilon_0} \frac{\mathbf{d} \cdot \mathbf{x}}{|\mathbf{x}|^3} + \mathcal{O}(|\mathbf{d}|^2). \end{aligned}$$

In the limit $|\mathbf{d}| \rightarrow 0$ with $q\mathbf{d}$ finite, we obtain a point dipole with electric dipole moment $\mathbf{p} = q\mathbf{d}$ and potential

$$\Phi(\mathbf{x}) = \frac{\mathbf{p} \cdot \mathbf{x}}{4\pi\epsilon_0 |\mathbf{x}|^3}$$

and electric field

$$\mathbf{E} = -\nabla\Phi = \frac{3(\mathbf{p} \cdot \mathbf{x})\mathbf{x} - |\mathbf{x}|^2\mathbf{p}}{4\pi\epsilon_0 |\mathbf{x}|^5}.$$

In spherical polar coordinates aligned with $\mathbf{p} = p\mathbf{e}_z$,

$$\begin{aligned} \Phi &= \frac{p \cos \theta}{4\pi\epsilon_0 r^2} & E_r &= -\frac{\partial\Phi}{\partial r} = \frac{2p \cos \theta}{4\pi\epsilon_0 r^3} \\ & & E_\theta &= -\frac{1}{r} \frac{\partial\Phi}{\partial \theta} = \frac{p \sin \theta}{4\pi\epsilon_0 r^3} \\ & & E_\phi &= 0 \end{aligned}$$

Note that Φ and $\mathbf{E}$ are not spherically symmetric. Also, they decrease more rapidly with r than for a point charge. A point dipole $\mathbf{p}$ at the origin corresponds to $\rho(\mathbf{x}) = -(\mathbf{p} \cdot \nabla)\delta(\mathbf{x})$ and

$$\Phi(\mathbf{x}) = (\mathbf{p} \cdot \nabla) \frac{1}{4\pi\epsilon_0 |\mathbf{x}|}.$$

2.2.3 Field lines and equipotentials

Electric field lines are the integral curves of $\mathbf{E}$, being tangent to $\mathbf{E}$ everywhere. Since $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, field lines begin on positive charges and end on negative charges. In electrostatics, $\mathbf{E} = -\nabla\Phi$ so field lines are perpendicular to the equipotential surfaces $\Phi = \text{constant}$.

2.2.4 Dipole in an external field

Consider a dipole $\mathbf{p}$ in an external electric field $\mathbf{E} = -\nabla\Phi$ generated by distant charges. With $-q$ at $\mathbf{x}$ and $+q$ at $\mathbf{x} + \mathbf{d}$, the potential energy of the dipole due to the external field is

$$U = -q\Phi(\mathbf{x}) + q\Phi(\mathbf{x} + \mathbf{d}) = q(\mathbf{d} \cdot \nabla)\Phi(\mathbf{x}) + \mathcal{O}(|\mathbf{d}|^2).$$

In the limit of a point dipole, $U = \mathbf{p} \cdot \nabla\Phi = -\mathbf{p} \cdot \mathbf{E}$ and is minimised when $\mathbf{p}$ is aligned with $\mathbf{E}$.

2.2.5 Multipole expansion

For a general charge distribution $\rho(\mathbf{x})$ confined to a ball $V = \{|\mathbf{x}| < R\}$,

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3\mathbf{x}'.$$

To get the external potential at $|\mathbf{x}| > R$, we expand

$$\begin{aligned} \frac{1}{|\mathbf{x} - \mathbf{x}'|} &= \frac{1}{r} - (\mathbf{x}' \cdot \nabla) \frac{1}{r} + \frac{1}{2} (\mathbf{x}' \cdot \nabla)^2 \frac{1}{r} + \mathcal{O}(|\mathbf{x}'|^3) \\ &= \frac{1}{r} \left[1 + \frac{\mathbf{x}' \cdot \mathbf{x}}{r^2} + \frac{3(\mathbf{x}' \cdot \mathbf{x})^2 - |\mathbf{x}'|^2 |\mathbf{x}|^2}{2r^4} + \mathcal{O}(R^3/r^3) \right]. \end{aligned}$$

This leads to the multipole expansion of the potential

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\mathbf{p} \cdot \mathbf{x}}{r^3} + \frac{1}{2} \frac{Q_{ij} x_i x_j}{r^5} + \dots \right)$$

which involves the first three multipole moments. The total charge $Q = \int_V \rho(\mathbf{x}) d^3\mathbf{x}$ is a scalar. The electric dipole moment $\mathbf{p} = \int_V \mathbf{x} \rho(\mathbf{x}) d^3\mathbf{x}$ is a vector. The electric quadrupole moment $Q_{ij} = \int_V (3x_i x_j - |\mathbf{x}|^2 \delta_{ij}) \rho(\mathbf{x}) d^3\mathbf{x}$ is a traceless symmetric second order tensor.

For $r \gg R$, Φ and $\mathbf{E}$ look increasingly like those of a point charge Q , unless $Q = 0$, in which case they look like those of a point dipole, unless $\mathbf{p} = \mathbf{0}$, and so on.

2.3 Electrostatic energy

Assume $\Phi = 0$ at infinity. The work done against the electric force $\mathbf{F} = q\mathbf{E}$ in bringing a particle of charge q from infinity to $\mathbf{x}$ is

$$- \int_{\infty}^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{x} = q \int_{\infty}^{\mathbf{x}} \nabla \Phi \cdot d\mathbf{x} = q\Phi(\mathbf{x}).$$

Consider assembling a configuration of N point charges one by one. Particle i of charge q_i is brought from infinity to $\mathbf{x}_i$ while the previous particles remain fixed. For the first particle, no work is involved so $W_1 = 0$. For the other particles,

$$\begin{aligned} W_2 &= q_2 \frac{q_1}{4\pi\epsilon_0 |\mathbf{x}_2 - \mathbf{x}_1|} \\ W_3 &= q_3 \left(\frac{q_1}{4\pi\epsilon_0 |\mathbf{x}_3 - \mathbf{x}_1|} + \frac{q_2}{4\pi\epsilon_0 |\mathbf{x}_3 - \mathbf{x}_2|} \right) \end{aligned} \quad \text{and so on.}$$

The total work done is

$$\begin{aligned} U &= \sum_{i=1}^N W_i = \sum_{i=1}^N \sum_{j=1}^{i-1} \frac{q_i q_j}{4\pi\epsilon_0 |\mathbf{x}_i - \mathbf{x}_j|} \\ &= \frac{1}{2} \sum_{i \neq j} \frac{q_i q_j}{4\pi\epsilon_0 |\mathbf{x}_i - \mathbf{x}_j|} \\ &= \frac{1}{2} \sum_{i=1}^N q_i \Phi(\mathbf{x}_i). \end{aligned}$$

We can generalise to a continuous charge distribution $\rho(\mathbf{x})$ occupying a finite volume V .

$$U = \frac{1}{2} \int_V \rho(\mathbf{x}) \Phi(\mathbf{x}) d^3\mathbf{x} = \frac{1}{2} \int_V \rho \Phi dV$$

Using (M1), we have

$$\begin{aligned} U &= \frac{1}{2} \int_V (\epsilon_0 \nabla \cdot \mathbf{E}) \Phi \, dV = \frac{\epsilon_0}{2} \int_V [\nabla \cdot (\Phi \mathbf{E}) - \mathbf{E} \cdot \nabla \Phi] \, dV \\ &= \frac{\epsilon_0}{2} \int_S \Phi \mathbf{E} \cdot d\mathbf{S} + \int_V \frac{\epsilon_0 |\mathbf{E}|^2}{2} \, dV. \end{aligned}$$

Let $S = \partial V$ be a sphere of radius $R \rightarrow \infty$. On S , we have $\Phi = \mathcal{O}(R^{-1})$ and $\mathbf{E} = \mathcal{O}(R^{-2})$. The area of S is $\mathcal{O}(R^2)$ so the first integral is $\mathcal{O}(R^{-1})$ and tends to zero as $R \rightarrow \infty$. Hence,

2.5 – Electrostatic energy

$$U = \int_{\mathbb{R}^3} \frac{\epsilon_0 |\mathbf{E}|^2}{2} \, dV$$

This implies that energy is stored in the electric field even in a vacuum. Any of these expressions for U suggests that the self-energy of a point charge is infinite. We can discard this as it is unchanging and causes no force.

2.4 Conductors

In a conductor (such as a metal), some charges (usually electrons) can move freely. In electrostatics, we require

$$\mathbf{E} = \mathbf{0}$$

$$\Phi = \text{constant}$$

inside a conductor, hence $\rho = 0$. Otherwise free charges would move in response to the electric force and a current would flow. A surface charge density σ can exist on the surface of a conductor, which is an equipotential. Taking $\mathbf{n}$ to point out of the conductor, the condition $[\mathbf{n} \cdot \mathbf{E}] = \sigma/\epsilon_0$ implies

$$\mathbf{n} \cdot \mathbf{E} = \frac{\sigma}{\epsilon_0}$$

immediately outside the conductor. The constant potential of a conductor can be set by connecting it to a battery or another conductor. An earthed or grounded conductor is connected to the ground and is usually taken as $\Phi = 0$.

To find $\Phi(\mathbf{x})$ and $\mathbf{E}(\mathbf{x})$ due to a charge distribution $\rho(\mathbf{x})$ in the presence of conductors with surfaces S_i and potentials Φ_i , we need to solve Poisson's equation

$$-\nabla^2 \Phi = \frac{\rho}{\epsilon_0}$$

with Dirichlet boundary conditions $\Phi = \Phi_i$ on S_i . The solution depends linearly on ρ and $\{\Phi_i\}$.

Example 2.1

Consider a point charge q at position $(0, 0, h)$ in a half-space $z > 0$ bounded by an earthed conducting wall. That is, $\Phi = 0$ at $z = 0$. By the method of images, the solution in $z > 0$ is identical to that of a dipole, with image charge $-q$ at $(0, 0, -h)$. The wall coincides with an equipotential of the dipole. The induced surface charge density on the wall can be worked out from

$$\frac{\sigma}{\epsilon_0} = \mathbf{n} \cdot \mathbf{E} = E_z = -\frac{2qh}{4\pi\epsilon_0(r^2 + h^2)^{3/2}}$$

where $r = (x^2 + y^2)^{1/2}$ is the distance from the origin. The total induced charge is

$$\int_0^\infty \sigma 2\pi r \, dr = -qh \int_0^\infty \frac{r}{(r^2 + h^2)^{3/2}} \, dr = -q$$

and is equal to the image charge.

A simple capacitor consists of two separated conductors carrying charges $\pm Q$. If the potential difference or voltage between them is V , then the capacitance is defined by

2.6 – Capacitance

$$C = \frac{Q}{V}$$

and depends only on the geometry because Φ depends linearly on Q .

Example 2.2

Consider two infinite parallel plates separated by a distance d . Let the plate surfaces at $z = 0$ and $z = d$ have surface charge densities $\pm\sigma$. Then, $\mathbf{E} = E\mathbf{e}_z$ with $E = \sigma/\epsilon_0$ for $0 < z < d$ and $\mathbf{E} = \mathbf{0}$ elsewhere. So, $\Phi = -Ez + \text{const.}$ and $V = Ed$. The same solution holds approximately for parallel plates of area $A \gg d^2$ if end effects are neglected. Hence,

$$C = \frac{Q}{V} \approx \frac{\sigma A}{Ed} \approx \frac{\epsilon_0 A}{d}.$$

The electrostatic energy stored in the capacitor is

$$U = \int \frac{\epsilon_0 |\mathbf{E}|^2}{2} dV = \frac{\epsilon_0 E^2}{2} Ad = \frac{1}{2} CV^2.$$

In general,

$$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

for any capacitor. The work done in moving an element of charge δQ from one plate to another is $\delta W = V \delta Q$, so the total work done is

$$\int_0^Q \frac{Q'}{C} dQ' = \frac{Q^2}{2C}.$$

Alternatively, we could use

$$U = \frac{1}{2} \int \rho \Phi dV = \frac{1}{2} Q \Phi_+ - \frac{1}{2} Q \Phi_- = \frac{1}{2} QV.$$

3 Magnetostatics

Magnetostatics is the study of the magnetic field generated by a stationary current distribution.

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

$$\nabla \cdot \mathbf{B} = 0$$

(M4') implies $\nabla \cdot \mathbf{J} = 0$ which is the time-independent equation of charge conservation.

3.1 Ampere's law

Consider a closed curve C that is the boundary of an open surface S . Integrate (M4') over S and apply Stokes's theorem to obtain

3.1 – Ampere's law

$$\int_C \mathbf{B} \cdot d\mathbf{x} = \mu_0 I$$

where $I = \int_S \mathbf{J} \cdot d\mathbf{S}$ is the total current through S .

Since $\nabla \cdot \mathbf{J} = 0$, the same current I flows through any open surface S such that $\partial S = C$. Ampere's law is the integral version of (M4') and is valid provided that $\partial \mathbf{E} / \partial t = 0$. We can interpret this as circulation of the magnetic field around a loop $\propto$ total current through the loop. In special situations, we can use Ampere's law together with symmetry to deduce $\mathbf{B}$ from $\mathbf{J}$.

A cylindrically symmetric situation could involve an axial current distribution $J_z(r)\mathbf{e}_z$ or an azimuthal current distribution $J_\phi(r)\mathbf{e}_\phi$ or a combination. Note that $\nabla \cdot \mathbf{J} = 0$ excludes a radial current. The same applies to $\mathbf{B}$. The curl in (M4') implies that B_ϕ is linearly related to J_z and B_z is linearly related to J_ϕ .

Example 3.1 – Long straight wire

Consider a cylindrical wire of radius R carrying a total current I parallel to its axis. To find $B_\phi(r)$ generated by $J_z(r)$, we apply Ampere's law to a circle C of radius r with S being a disc. If $r > R$, then

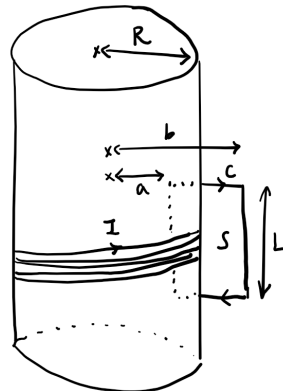
$$\int_C \mathbf{B} \cdot d\mathbf{x} = B_\phi(r) \int_C \mathbf{e}_\phi \cdot d\mathbf{x} = B_\phi(r) \int_C dl = 2\pi r B_\phi(r) = \mu_0 I$$

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r} \mathbf{e}_\phi.$$

Example 3.2 – Solenoid

A thin wire is coiled around a cylindrical tube of radius R . An ideal solenoid is infinitely long and tightly wound, having cylindrical symmetry and purely azimuthal current. The wire carries current I and has N turns per unit length of the tube. To find $B_z(r)$ generated by $J_\phi(r)$, we apply Ampere's law to a rectangular loop C . Taking $a < b < R$ or $R < a < b$ gives $L(B_z(a) - B_z(b)) = 0$. Taking $a < R < b$ gives $L(B_z(a) - B_z(b)) = \mu_0 NLI$. Assuming that $B_z(r) \rightarrow 0$ as $r \rightarrow \infty$, we deduce that

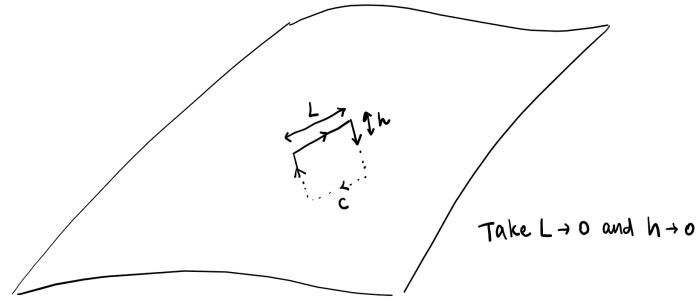
$$B_z(r) = \begin{cases} \mu_0 NI & \text{if } r < R \\ 0 & \text{if } r > R. \end{cases}$$



The ideal solenoid is an example of a surface current. It has the form $J_\phi(r) = K_\phi \delta(r - R)$ with $K_\phi = NI$ for a solenoid. Generally, a surface current density $\mathbf{K}$ produces a discontinuity in the tangential magnetic field

$$[\mathbf{n} \times \mathbf{B}] = \mu_0 \mathbf{K}.$$

This follows from Ampere's law applied to the following.



(M2) implies that the normal component is continuous so

$$[\mathbf{n} \cdot \mathbf{B}] = 0.$$

3.2 The magnetic vector potential

(M2) implies that $\mathbf{B}$ can be written in terms of a

3.2 – Magnetic vector potential $\mathbf{A}(\mathbf{x})$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Note that $\mathbf{A}$ is not unique. If we make a gauge transformation, replacing $\mathbf{A}$ with $\tilde{\mathbf{A}} = \mathbf{A} + \nabla\chi$ where $\chi(\mathbf{x})$ is an arbitrary scalar field, then $\mathbf{B}$ is unchanged. A convenient gauge for many calculations is the Coulomb gauge in which

$$\nabla \cdot \mathbf{A} = 0.$$

We can assume this condition without loss of generality. If $\nabla \cdot \mathbf{A} \neq 0$, then we can make a gauge transformation such that $\nabla \cdot \tilde{\mathbf{A}} = 0$ by choosing χ to be the solution of Poisson's equation $-\nabla^2\chi = \nabla \cdot \mathbf{A}$.

In terms of $\mathbf{A}$, (M4') becomes $\nabla \times (\nabla \times \mathbf{A}) = \mu_0\mathbf{J}$. Using the identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2\mathbf{A}$ and assuming Coulomb gauge, we obtain Poisson's equation in vector form

$$-\nabla^2\mathbf{A} = \mu_0\mathbf{J}.$$

3.3 The Biot-Savart law

The solution of Poisson's equation is (integrating over all space and assuming decay at infinity)

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3\mathbf{x}'.$$

We should check that the solution satisfies the assumed Coulomb gauge condition.

$$\begin{aligned}
\nabla \cdot \mathbf{A}(\mathbf{x}) &= \frac{\mu_0}{4\pi} \int_V \nabla \cdot \left(\frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \right) d^3\mathbf{x}' \\
&= \frac{\mu_0}{4\pi} \int_V \mathbf{J}(\mathbf{x}') \cdot \nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3\mathbf{x}' \\
&= -\frac{\mu_0}{4\pi} \int_V \mathbf{J}(\mathbf{x}') \cdot \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) d^3\mathbf{x}' \\
&= -\frac{\mu_0}{4\pi} \int_V \nabla' \cdot \left(\frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \right) d^3\mathbf{x}' && \text{using } \nabla \cdot \mathbf{J} = 0 \\
&= -\frac{\mu_0}{4\pi} \int_{\partial V} \frac{\mathbf{J}(\mathbf{x}') \cdot d\mathbf{S}'}{|\mathbf{x} - \mathbf{x}'|}
\end{aligned}$$

where ∇' represents differentiation with respect to $\mathbf{x}'$. Thus $\nabla \cdot \mathbf{A} = 0$ as assumed, if the current is contained in some finite volume and we take V to be at least as large, or if $\mathbf{J}$ decays sufficiently quickly as $|\mathbf{x}| \rightarrow \infty$. To find the magnetic field, we use $\mathbf{B} = \nabla \times \mathbf{A}$.

3.3 – Biot-Savart law

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int_{\mathbb{R}^3} \frac{\mathbf{J}(\mathbf{x}') \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} d^3\mathbf{x}'$$

This gives the magnetic field generated by a stationary current distribution.

A special case is when the current is restricted to a thin wire in the form of a curve C . Then the current element $\mathbf{J} d^3\mathbf{x}$ can be replaced by $I d\mathbf{x}$, where I is the current in the wire and $d\mathbf{x}$ is the vector line element parallel to the wire. Charge conservation implies that I is constant along the wire. The result is

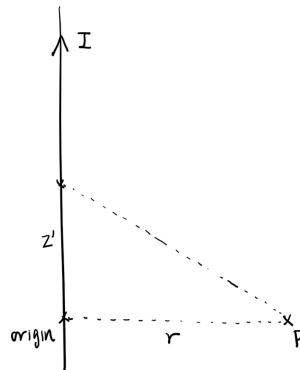
$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0 I}{4\pi} \int_C \frac{d\mathbf{x}' \times (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3}.$$

The thin wire density can be represented as $\mathbf{J}(\mathbf{x}) = I \int_C \delta(\mathbf{x} - \mathbf{x}') d\mathbf{x}'$, which gives the above result when substituted into the Biot-Savart law.

Charge conservation takes the form

$$\begin{aligned}
\nabla \cdot \mathbf{J}(\mathbf{x}) &= I \int_C \nabla \delta(\mathbf{x} - \mathbf{x}') \cdot d\mathbf{x}' \\
&= -I \int_C \nabla' \delta(\mathbf{x} - \mathbf{x}') \cdot d\mathbf{x}' \\
&= -I[\delta(\mathbf{x} - \mathbf{x}_2) - \delta(\mathbf{x} - \mathbf{x}_1)]
\end{aligned}$$

where C runs from $\mathbf{x}_1$ to $\mathbf{x}_2$. If C is closed, then $\mathbf{x}_1 = \mathbf{x}_2$ and $\nabla \cdot \mathbf{J} = 0$ as expected. If C is infinite, then $\nabla \cdot \mathbf{J} = 0$ for any finite $\mathbf{x}$.



We can check that the Biot-Savart law gives the same result as Ampere's law for a long straight thin wire along the z -axis in cylindrical polars. We have $\mathbf{x} = r\mathbf{e}_r$ and $\mathbf{x}' = z'\mathbf{e}_z$. So, $\mathbf{x} - \mathbf{x}' = r\mathbf{e}_r - z'\mathbf{e}_z$ and $d\mathbf{x}' = dz'\mathbf{e}_z$. This gives

$$\begin{aligned}\mathbf{B}(\mathbf{x}) &= \frac{\mu_0 I}{4\pi} \mathbf{e}_\phi \int_{-\infty}^{\infty} \frac{r dz'}{(r^2 + (z')^2)^{3/2}} \\ &= \frac{\mu_0 I}{4\pi} \mathbf{e}_\phi \left[\frac{z'}{r(r^2 + (z')^2)^{3/2}} \right]_{-\infty}^{\infty} \\ &= \frac{\mu_0 I}{2\pi r} \mathbf{e}_\phi.\end{aligned}$$

3.4 Magnetic dipoles

For a general current distribution $\mathbf{J}(\mathbf{x})$ confined to a ball $V = \{|\mathbf{x}| < R\}$,

$$\mathbf{A}(\mathbf{x}) = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3\mathbf{x}'.$$

The external field for $|\mathbf{x}| = r > R$ can be evaluated by expanding

$$\frac{1}{|\mathbf{x} - \mathbf{x}'|} = \frac{1}{r} \left(1 + \frac{\mathbf{x}' \cdot \mathbf{x}}{r^2} + \mathcal{O}(R^2/r^2) \right),$$

leading to a multipole expansion as before. We need to calculate the moments of the current distribution. Since $\mathbf{J} = \mathbf{0}$ on ∂V and $\nabla \cdot \mathbf{J} = 0$, the divergence theorem implies

$$\begin{aligned}0 &= \int_{\partial V} x_i J_j dS_j = \int_V \partial_j (x_i J_j) d^3\mathbf{x} \\ &= \int_V (\delta_{ij} J_j + x_i \partial_j J_j) d^3\mathbf{x} \\ &= \int_V J_i d^3\mathbf{x}\end{aligned}$$

so the zeroth moment vanishes. Similarly,

$$\begin{aligned}0 &= \int_{\partial V} x_i x_j J_k dS_k = \int_V \partial_k (x_i x_j J_k) d^3\mathbf{x} \\ &= \int_V (\delta_{ik} x_j J_k + x_i \delta_{jk} J_k + x_i x_j \partial_k J_k) d^3\mathbf{x} \\ &= \int_V x_j J_i d^3\mathbf{x} + \int_V x_i J_j d^3\mathbf{x}\end{aligned}$$

so the first moment is an antisymmetric matrix. The magnetic dipole moment is

$$\mathbf{m} = \frac{1}{2} \int_V \mathbf{x} \times \mathbf{J} d^3\mathbf{x}.$$

Or in suffix notation,

$$m_i = \frac{1}{2} \epsilon_{ijk} \int_V x_j J_k d^3\mathbf{x}.$$

It is a vector related to the antisymmetric matrix by

$$\int_V x_i J_j d^3\mathbf{x} = \epsilon_{ijk} m_k.$$

Returning to the multipole expansion for $\mathbf{A}$, we have

$$A_i(\mathbf{x}) = \frac{\mu_0}{4\pi|\mathbf{x}|} \left(\int_V J_i(\mathbf{x}') d^3\mathbf{x}' + \frac{x_j}{|\mathbf{x}|^2} \int_V x'_j J_i(\mathbf{x}') d^3\mathbf{x}' + \dots \right) = \frac{\mu_0}{4\pi|\mathbf{x}|} \left(0 + \frac{x_j \epsilon_{jik} m_k}{|\mathbf{x}|^2} + \dots \right).$$

The leading approximation is therefore

$$\mathbf{A}(\mathbf{x}) \approx \mathbf{A}_{\text{dipole}}(\mathbf{x}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{x}}{|\mathbf{x}|^3}$$

which is the vector potential due to a point dipole $\mathbf{m}$ at the origin. The corresponding magnetic field is

$$\mathbf{B}_{\text{dipole}} = \nabla \times \mathbf{A}_{\text{dipole}} = \frac{\mu_0}{4\pi} \frac{3(\mathbf{m} \cdot \mathbf{x})\mathbf{x} - |\mathbf{x}|^2 \mathbf{m}}{|\mathbf{x}|^5}.$$

A point dipole $\mathbf{m}$ at the origin corresponds to the current density and vector potential

$$\mathbf{J} = \nabla \times (\mathbf{m} \delta(\mathbf{x})) \quad \mathbf{A} = \nabla \times \left(\frac{\mu_0 \mathbf{m}}{4\pi |\mathbf{x}|} \right).$$

The magnetic dipole moment of a thin wire carrying current I around a closed curve C is

$$\mathbf{m} = \frac{I}{2} \int_C \mathbf{x} \times d\mathbf{x}.$$

To evaluate this, let $\mathbf{a}$ be any constant vector. Then, by Stokes' theorem,

$$\begin{aligned} \mathbf{a} \cdot \mathbf{m} &= \frac{I}{2} \int_C \mathbf{a} \cdot (\mathbf{x} \times d\mathbf{x}) \\ &= \frac{I}{2} \int_C (\mathbf{a} \times \mathbf{x}) \cdot d\mathbf{x} \\ &= \frac{I}{2} \int_S (\nabla \times (\mathbf{a} \times \mathbf{x})) \cdot d\mathbf{S} = I \int_S \mathbf{a} \cdot d\mathbf{S} \end{aligned}$$

where S is an open surface with boundary C and we used

$$\nabla \times (\mathbf{a} \times \mathbf{x}) = \mathbf{x} \cdot \nabla \mathbf{a} - \mathbf{a} \cdot \nabla \mathbf{x} + (\nabla \cdot \mathbf{x}) \mathbf{a} - (\nabla \cdot \mathbf{a}) \mathbf{x} = \mathbf{0} - \mathbf{a} + 3\mathbf{a} - \mathbf{0} = 2\mathbf{a}.$$

Since $\mathbf{a}$ is arbitrary, we have $\mathbf{m} = IS$ where $\mathbf{S} = \int_S d\mathbf{S}$ is the vector area of S (which depends only on C).

Example 3.3 – Circular loop

The simplest example of this would be a circular loop $x^2 + y^2 = a^2$. The magnetic dipole moment would be $\mathbf{m} = I\pi a^2 \mathbf{e}_z$. On the z -axis, the dipole approximation gives

$$B_z = \frac{\mu_0}{4\pi} \left(\frac{3m_z z^2 - z^2 m_z}{|z|^5} \right) = \frac{\mu_0 I a^2}{2|z|^3}$$

while the exact solution is

$$B_z = \frac{\mu_0 I a^2}{2(z^2 + a^2)^{3/2}}.$$

The magnetic field lines are the integral curves of $\mathbf{B}$. Since $\nabla \cdot \mathbf{B} = 0$, they are continuous.

A bar magnet has north and south poles, and a dipole moment. This comes from the superposition of aligned dipoles on the atomic scale. Atoms contain electrons, which are spinning charged particles, with a magnetic dipole moment. A classical model of a particle is a spinning charged sphere, which is a current loop with a magnetic dipole moment proportional to its charge and spin. As far as we know, there are no magnetic charges or monopoles.

The Earth also acts as a magnet. The liquid iron outer core of the Earth is a conducting fluid in convective motion, and supports electric currents that generate a magnetic field. At the Earth's surface, this resembles a dipole field.

3.5 Magnetic forces

The Lorentz force on a particle of charge q_i at position $\mathbf{x}_i(t)$ is $q_i(\mathbf{E} + \dot{\mathbf{x}}_i \times \mathbf{B})$ where $\mathbf{E}$ and $\mathbf{B}$ are evaluated at $\mathbf{x}_i(t)$. In the limit of continuous charge and current densities, the Lorentz force per unit volume is

$$\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}.$$

We can recover the discrete version by substituting $\rho = \sum_i q_i \delta(\mathbf{x} - \mathbf{x}_i(t))$ and $\mathbf{J} = \sum_i q_i \dot{\mathbf{x}}_i(t) \delta(\mathbf{x} - \mathbf{x}_i(t))$.

Consider two or more thin wires with current I_i along curves C_i . The total magnetic field is $\mathbf{B} = \sum_i \mathbf{B}_i$ where

$$\mathbf{B}_i(\mathbf{x}) = \frac{\mu_0 I_i}{4\pi} \int_{C_i} \frac{d\mathbf{x}_i \times (\mathbf{x} - \mathbf{x}_i)}{|\mathbf{x} - \mathbf{x}_i|^3}$$

is the magnetic field due to wire i . The current density is $\mathbf{J} = \sum_i \mathbf{J}_i$ where

$$\mathbf{J}_i(\mathbf{x}) = I_i \int_{C_i} \delta(\mathbf{x} - \mathbf{x}_i) d\mathbf{x}_i.$$

The total magnetic force acting on a volume V is $\mathbf{F} = \int_V \mathbf{J} \times \mathbf{B} dV$. The force acting on wire i is

$$\mathbf{F}_i = \int \mathbf{J}_i(\mathbf{x}) \times \mathbf{B}(\mathbf{x}) d^3\mathbf{x} = I_i \int_{C_i} d\mathbf{x}_i \times \mathbf{B}(\mathbf{x}_i).$$

Since $\mathbf{B} = \sum_i \mathbf{B}_i$, we have $\mathbf{F}_i = \sum_j \mathbf{F}_{ij}$ where $\mathbf{F}_{ij} = I_i \int_{C_i} d\mathbf{x}_i \times \mathbf{B}_j(\mathbf{x}_i)$ is the force on wire i due to wire j . Using the Biot-Savart law,

$$\mathbf{F}_{ij} = \frac{\mu_0 I_i I_j}{4\pi} \int_{C_i} \int_{C_j} d\mathbf{x}_i \times \left(\frac{d\mathbf{x}_j \times (\mathbf{x}_i - \mathbf{x}_j)}{|\mathbf{x}_i - \mathbf{x}_j|^3} \right).$$

This can be written in a manifestly antisymmetric way that shows that $\mathbf{F}_{ji} = -\mathbf{F}_{ij}$ as expected from Newton's third law. The self-force $\mathbf{F}_{ii}$ vanishes, although the thin-wire integral is singular, we can consider thick wires.

Example 3.4

Consider two infinitely-long parallel thin wires separated by a distance r . Use cylindrical polars centred on wire 2. We have $\mathbf{B}_2 = \mu_0 I_2 / (2\pi r) \mathbf{e}_\phi$ so

$$\mathbf{F}_{12} = I_1 \int_{-\infty}^{\infty} dz \mathbf{e}_z \times \mathbf{B}_2.$$

The total force is infinite. The force per unit length is

$$I_1 \mathbf{e}_z \times \mathbf{B}_2 = -\frac{\mu_0 I_1 I_2}{2\pi r} \mathbf{e}_r.$$

This is directed towards wire 2 if $I_1 I_2 > 0$. So the force is attractive if the currents are aligned, and is repulsive otherwise.

Consider a localised current distribution (current loop) confined to a ball $V = \{|\mathbf{x}| < R\}$. We place this in an external magnetic field $\mathbf{B}(\mathbf{x})$ that varies slowly over the lengthscale R . The magnetic torque about the origin on the current loop is

$$\boldsymbol{\tau} = \int_V \mathbf{x} \times (\mathbf{J}(\mathbf{x}) \times \mathbf{B}(\mathbf{x})) d^3\mathbf{x} = \int_V [(\mathbf{x} \cdot \mathbf{B}(\mathbf{x}))\mathbf{J}(\mathbf{x}) - (\mathbf{x} \cdot \mathbf{J}(\mathbf{x}))\mathbf{B}(\mathbf{x})] d^3\mathbf{x}.$$

Within V , $\mathbf{B}(\mathbf{x})$ can be expanded as a Taylor series, $B_i(\mathbf{x}) = B_i(\mathbf{0}) + x_j \partial_j B_i(\mathbf{0}) + \dots$. Retaining only the zeroth-order term (a uniform field), we have

$$\tau_i = B_j(\mathbf{0}) \int_V x_j J_i d^3\mathbf{x} - B_i(\mathbf{0}) \int_V x_j J_j d^3\mathbf{x}.$$

Recall the first moments of the current distribution $\int_V x_i J_j d^3\mathbf{x} = \epsilon_{ijk} m_k$. Thus, $\tau_i = B_j(\mathbf{0}) \epsilon_{jik} m_k$. In general,

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$$

where $\mathbf{B}$ is evaluated at the dipole's location and τ is measured about this point.

For the force, we need to go to the first-order of the Taylor expansion of $\mathbf{B}$.

$$\begin{aligned} \mathbf{F} &= \int_V \mathbf{J}(\mathbf{x}) \times \mathbf{B}(\mathbf{x}) d^3\mathbf{x} \\ F_i &\approx \int_V \epsilon_{ijk} J_j(\mathbf{x}) (B_k(\mathbf{0}) + x_l \partial_l B_k(\mathbf{0})) d^3\mathbf{x} \\ &= \epsilon_{ijk} B_k(\mathbf{0}) \int_V J_j d^3\mathbf{x} + \epsilon_{ijk} \partial_l B_k(\mathbf{0}) \int_V x_l J_j d^3\mathbf{x} \\ &= 0 + \epsilon_{ijk} \partial_l B_k(\mathbf{0}) \epsilon_{l j n} m_n \\ &= \partial_i B_k(\mathbf{0}) m_k - \partial_k B_k(\mathbf{0}) m_i \\ &= \partial_i (m_k B_k)(\mathbf{0}) \end{aligned}$$

since $\nabla \cdot \mathbf{B} = 0$. In general, $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$. This can also be written as $\mathbf{F} = -\nabla U$ where $U = -\mathbf{m} \cdot \mathbf{B}$ is the potential energy of a magnetic dipole in an external field. As in the electric case, this is minimised when $\mathbf{m}$ is aligned with $\mathbf{B}$.

4 Electrodynamics

4.1 Faraday's law of induction

Maxwell's third equation

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

implies that a time-dependent magnetic field must be accompanied by an electric field. This can induce a current to flow in a conductor, a process known as electromagnetic induction.

4.1.1 Faraday's law for a static circuit

Consider a closed curve C that is the boundary of a time-independent open surface S . Integrate (M3) over S and use Stokes' theorem.

$$\int_C \mathbf{E} \cdot d\mathbf{x} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = - \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}.$$

Rewriting, we get

4.1 – Faraday's law of induction for a static circuit

$$\mathcal{E} = -\frac{d\mathcal{F}}{dt}$$

where $\mathcal{E} = \int_C \mathbf{E} \cdot d\mathbf{x}$ is the electromagnetic force (emf) around C and $\mathcal{F} = \int_S \mathbf{B} \cdot d\mathbf{S}$ is the magnetic flux through S .

Since $\nabla \cdot \mathbf{B} = 0$, the flux $\mathcal{F}$ is the same for any S such that $\partial S = C$, so it can be regarded as the magnetic flux through C . Using $\mathbf{B} = \nabla \times \mathbf{A}$ and Stokes' theorem, we can write $\mathcal{F} = \int_C \mathbf{A} \cdot d\mathbf{x}$, which is invariant under a gauge transformation.

The emf is not actually a force. It is the line integral of the Lorentz force on a particle of unit charge confined

to C .

$$\frac{1}{q} \int_C \mathbf{F} \cdot d\mathbf{x} = \int_C (\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}) \cdot d\mathbf{x} = \int_C \mathbf{E} \cdot d\mathbf{x} = \mathcal{E}$$

since $\dot{\mathbf{x}}$ is tangent to C for a particle confined to a time-independent curve C . We will see later that if C coincides with a thin wire of resistance R , then the current induced in the wire is $I = \mathcal{E}/R$.

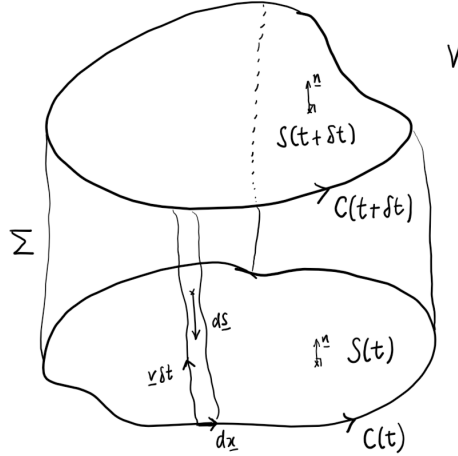
There are several ways in which the magnetic flux through C could change in time. A magnet could be moved near C . A current-carrying circuit could be moved near C . The current in a nearby circuit could be changed. All these will induce an emf around C and cause a current to flow.

4.1.2 Faraday's law for a moving circuit

Now let $C(t)$ be a time-dependent closed curve that is the boundary of an open surface $S(t)$. How does the magnetic flux $\mathcal{F}$ through S change in time? We have

$$\begin{aligned} \mathcal{F}(t + \delta t) - \mathcal{F}(t) &= \int_{S(t+\delta t)} \mathbf{B}(\mathbf{x}, t + \delta t) \cdot d\mathbf{S} - \int_{S(t)} \mathbf{B}(\mathbf{x}, t) \cdot d\mathbf{S} \\ &= \int_{S(t+\delta t)} \left(\mathbf{B}(\mathbf{x}, t) + \frac{\partial \mathbf{B}}{\partial t} \delta t + \mathcal{O}(\delta t^2) \right) \cdot d\mathbf{S} - \int_{S(t)} \mathbf{B}(\mathbf{x}, t) \cdot d\mathbf{S} \\ &= \int_{S(t+\delta t) - S(t)} \mathbf{B}(\mathbf{x}, t) \cdot d\mathbf{S} + \int_{S(t)} \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \delta t + \mathcal{O}(\delta t^2) \end{aligned}$$

since $S(t + \delta t) = S(t) + \mathcal{O}(\delta t)$. Let δV be the volume swept out by $S(t)$ in the time interval δt . Its boundary is the closed surface $S(t + \delta t) - S(t) + \Sigma$ where Σ is the surface swept out by $C(t)$ in time δt .



By (M2) and the divergence theorem,

$$0 = \int_{\partial V} (\nabla \cdot \mathbf{B}) dV = \int_{S(t+\delta t) - S(t)} \mathbf{B} \cdot d\mathbf{S} + \int_{\Sigma} \mathbf{B} \cdot d\mathbf{S}.$$

To evaluate the last term, parameterise C as $\mathbf{x} = \mathbf{x}(\lambda, t)$ where λ is a parameter around C . An element of C is $d\mathbf{x} = (\partial \mathbf{x} / \partial \lambda) d\lambda$ and has velocity $\mathbf{v} = \partial \mathbf{x} / \partial t$. In time δt , it sweeps out the vector area element $d\mathbf{S} = d\mathbf{x} \times (\mathbf{v} \delta t)$, which points out of ∂V as required. Thus,

$$\int_{\Sigma} \mathbf{B} \cdot d\mathbf{S} = \int_C \mathbf{B} \cdot (d\mathbf{x} \times \mathbf{v}) \delta t + \mathcal{O}(\delta t^2).$$

We then have

$$\mathcal{F}(t + \delta t) - \mathcal{F}(t) = - \int_C (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x} \delta t + \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \delta t + \mathcal{O}(\delta t^2).$$

Hence,

$$\begin{aligned}\frac{d\mathcal{F}}{dt} &= - \int_C (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x} + \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \\ &= - \int_C (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x} - \int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} \\ &= - \int_C (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x}.\end{aligned}$$

We recover Faraday's law $\mathcal{E} = -d\mathcal{F}/dt$ with the redefined emf

$$\mathcal{E} = \int_C (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x}.$$

This is again the line integral around C of the Lorentz force on a particle of unit charge confined to C , for which the perpendicular components of the particle's velocity $\dot{\mathbf{x}}$ must agree with those of the curve velocity $\mathbf{v}$.

4.2 – Lenz's law

The direction of the induced current is always such as to produce a magnetic field that opposes the change in flux that causes the emf.

Example 4.1

Consider a circular wire in the xy -plane. If B_z inside the loop increases over time, then $\mathcal{E} = -d\mathcal{F}/dt < 0$. This induces a clockwise current $I < 0$ that generates magnetic field $B_z < 0$ inside the loop. Hence the minus sign in Faraday's law. This avoids an unstable situation in which the flux grows indefinitely.

Definition 4.1 – Inductance

If a current I around a circuit C generates a magnetic field with flux $\mathcal{F}$, then the inductance L of the circuit is defined by

$$L = \frac{\mathcal{F}}{I}$$

and depends only on the geometry.

Example 4.2

Consider an ideal solenoid with cross-sectional area A and N turns per unit length. The uniform field $B = \mu_0 N I$ inside the solenoid produces a flux BA per turn, so the inductance per unit length of the solenoid is $\mu_0 N^2 A$.

We can show that the magnetic flux through a thin wire C_i due to a current I_j around another thin wire C_j is $\mathcal{F}_{ij} = L_{ij} I_j$ where the mutual inductance is

$$L_{ij} = \frac{\mu_0}{4\pi} \int_{C_i} \int_{C_j} \frac{d\mathbf{x}_i \cdot d\mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|} = L_{ji}.$$

When the current I around a circuit C is varied, an emf

$$\mathcal{E} = -\frac{d\mathcal{F}}{dt} = -L \frac{dI}{dt}$$

is induced. In a small time interval δt , a charge $\delta Q = I \delta t$ flows around C , and the work done on it by the Lorentz force is $\delta W = \mathcal{E} \delta Q = -LI(dI/dt) \delta t$. So, the rate at which work is done by the current on the electromagnetic field is

$$-\frac{dW}{dt} = LI \frac{dI}{dt} = \frac{d}{dt} \left(\frac{1}{2} L I^2 \right).$$

Consider reaching a magnetostatic state by building up the current from 0 to I . The energy stored is

$$U = \frac{1}{2}LI^2 = \frac{1}{2}I\mathcal{F} = \frac{1}{2}I \int_C \mathbf{A} \cdot d\mathbf{x} = \frac{1}{2} \int \mathbf{J} \cdot \mathbf{A} dV$$

analogous to $U = (1/2) \int \rho\Phi dV$ in electrostatics. Now, using (M3'), we have

$$U = \frac{1}{2\mu_0} \int (\nabla \times \mathbf{B}) \cdot \mathbf{A} dV.$$

Note that $(\nabla \times \mathbf{B}) \cdot \mathbf{A} = \nabla \cdot (\mathbf{B} \times \mathbf{A}) + \mathbf{B} \cdot (\nabla \times \mathbf{A})$. If we take the integral over all space, then the first term $\nabla \cdot (\mathbf{B} \times \mathbf{A})$ gives zero by the divergence theorem, since $|\mathbf{B}| = \mathcal{O}(|\mathbf{x}|^{-3})$ and $|\mathbf{A}| = \mathcal{O}(|\mathbf{x}|^{-2})$ as $|\mathbf{x}| \rightarrow \infty$ for a finite current distribution. Hence,

$$U = \int_{\mathbb{R}^3} \frac{|\mathbf{B}|^2}{2\mu_0} dV$$

is the magnetic energy stored in the magnetic field.

4.2 Ohm's law

In a stationary conductor,

$$\mathbf{J} = \sigma \mathbf{E}$$

where σ is the electrical conductivity. This is not a fundamental physical law but a constitutive relation, a macroscopic property of a material. The inverse relation $\mathbf{E} = \sigma^{-1}\mathbf{J}$ where σ^{-1} is the resistivity. This is usually denoted ρ but both ρ and σ conflict with the notation for charge densities, so we would not write $\rho = \sigma^{-1}$ here. In general, σ would be a symmetric tensor instead of a scalar. A perfect conductor corresponds to the limit $\sigma \rightarrow \infty$ so $\mathbf{E} = \mathbf{0}$. A perfect insulator corresponds to the limit $\sigma \rightarrow 0$ so $\mathbf{J} = \mathbf{0}$.

Consider a straight wire of length L in the direction of the unit vector $\mathbf{n}$. Assume it has uniform cross-sectional area A and conductivity σ . If the electric field is $\mathbf{E} = E\mathbf{n}$ where E is a constant, then $\mathbf{J} = \sigma E\mathbf{n}$ and the total current is $I = \sigma EA$. The potential difference along the wire is $V = \int \mathbf{E} \cdot d\mathbf{x} = EL = IL/(\sigma A)$. This gives

$$V = IR$$

where $R = L/\sigma A$ is the resistance of the wire.

Accompanying the resistance of a wire is Joule heating or Ohmic heating, which is the conversion of electromagnetic energy into heat at the rate $I^2 R$. If the voltage V is maintained by a battery, then $VI = I^2 R$ is the rate at which the emf of the battery does work to maintain the current I .

4.3 Time-dependent electric fields

4.3.1 Scalar and vector potentials

In electrodynamics, we can no longer write $\mathbf{E} = -\nabla\Phi$. But (M2) still allows us to write $\mathbf{B} = \nabla \times \mathbf{A}$ and (M3) then gives

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = \mathbf{0}.$$

This allows us to write

$$\mathbf{E} = -\nabla\Phi - \frac{\partial \mathbf{A}}{\partial t},$$

generalising the electrostatic expression. Under a time-dependent gauge transformation $\tilde{\mathbf{A}} = \mathbf{A} + \nabla\chi$ and $\tilde{\Phi} = \Phi - \partial\chi/\partial t$ where $\chi(\mathbf{x}, t)$ is any scalar field, then both $\mathbf{E}$ and $\mathbf{B}$ are unchanged.

4.3.2 The displacement current

In magnetostatics, we used Ampere's law $\int_C \mathbf{B} \cdot d\mathbf{x} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{S} = \mu_0 I$ or its differential form $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. For time-dependent situations,

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

contains an extra term, the displacement current. Why is it needed? Without it, we would have $\nabla \cdot \mathbf{J} = 0$, which describes charge conservation in a situation where ρ is constrained to remain constant. But suppose we place free particles of positive charge in some localised region. Repulsive Coulomb forces cause the particles to separate, implying $\nabla \cdot \mathbf{J} > 0$. We have seen that charge conservation in the correct form

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

follows from Maxwell's equation including the displacement current.

4.4 Electromagnetic waves

Consider freely evolving electric and magnetic fields in a vacuum, in the absence of charges and currents.

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0 & \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 & \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{aligned}$$

Eliminate $\mathbf{B}$ by taking $(\partial/\partial t)$ (M4) and substituting (M3).

$$\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = \nabla \times \frac{\partial \mathbf{B}}{\partial t} = -\nabla \times (\nabla \times \mathbf{E}) = \nabla^2 \mathbf{E}$$

where we used the identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$ and (M1). Alternatively, eliminate $\mathbf{E}$ by taking $(\partial/\partial t)$ (M3) and substituting (M4).

$$\frac{\partial^2 \mathbf{B}}{\partial t^2} = -\frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\nabla \times \frac{\partial \mathbf{E}}{\partial t} = -\frac{1}{\mu_0 \epsilon_0} \nabla \times (\nabla \times \mathbf{B}) = \frac{1}{\mu_0 \epsilon_0} \nabla^2 \mathbf{B}$$

where we used the same identity and (M2). So each Cartesian component of $\mathbf{E}$ and $\mathbf{B}$ satisfies the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

with wave speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

which is the speed of light in a vacuum. This is of course because light is an electromagnetic wave involving oscillations of $\mathbf{E}$ and $\mathbf{B}$. Depending on the wavelength, electromagnetic waves can be radio waves, microwaves, infrared, ultraviolet, X-rays, gamma rays, and so on.

4.4.1 Plane electromagnetic waves

Consider a plane wave in which $\mathbf{E}$ and $\mathbf{B}$ depend only on (x, t) and not on (y, z) . A simple example is $\mathbf{E} = E(x, t) \mathbf{e}_y$ where $E(x, t)$ satisfies the 1D wave equation

$$\frac{\partial^2 E}{\partial t^2} = c^2 \frac{\partial^2 E}{\partial x^2}.$$

The general solution $E(x, t) = f(x - ct) + g(x + ct)$ is the sum of a wave travelling without change of form in the $+x$ direction and another travelling in the $-x$ direction. What is the corresponding $\mathbf{B}$? We have

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} = -\frac{\partial E}{\partial x} \mathbf{e}_z = [-f'(x - ct) - g'(x + ct)] \mathbf{e}_z$$

and so $\mathbf{B} = B(x, t) \mathbf{e}_z$ with $B(x, t) = (1/c)[f(x - ct) - g(x + ct)]$. This also satisfies (M4).

Of particular importance is a monochromatic wave of a single angular frequency ω . For example, $E = E_0 \cos(kx - \omega t)$ and $B = (E_0/c) \cos(kx - \omega t)$, where E_0 is a constant amplitude and $k = \omega/c$ is the wavenumber, which is related to the wavelength λ by $k = 2\pi/\lambda$. The frequency is $\nu = \omega/(2\pi)$ and the period is $1/\nu = 2\pi/\omega$.

- (i) The angular frequency and wavenumber are related by the dispersion relation $\omega^2 = c^2 k^2$. That is, $\omega = \pm ck$.
- (ii) The oscillations of $\mathbf{E}$ and $\mathbf{B}$ are in phase but in orthogonal directions.
- (iii) The waves are transverse. That is, the oscillatory fields are orthogonal to the direction in which the wave varies and propagates.

Because Maxwell's equations are linear, electromagnetic waves of different amplitudes, frequencies and directions can be superposed. Physically, this is because light rays can pass through each other.

4.4.2 Polarisation

A more general approach to plane electromagnetic waves is to seek solutions of the form $\mathbf{E} = \text{Re}(\mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)})$ and $\mathbf{B} = \text{Re}(\mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)})$ where $\mathbf{E}_0$ and $\mathbf{B}_0$ are complex constant vector amplitudes, $\mathbf{k}$ is the real constant wavevector, and ω is the real constant angular frequency. The wavenumber is $k = |\mathbf{k}|$. The wave equation is satisfied by $\mathbf{E}$ and $\mathbf{B}$ if ω and k satisfy the dispersion relation $\omega^2 = c^2 k^2$. Since we are only taking the real part, different terms inside $\text{Re}(\)$ could describe the same physical field. The individual Maxwell equations reduce to algebraic conditions.

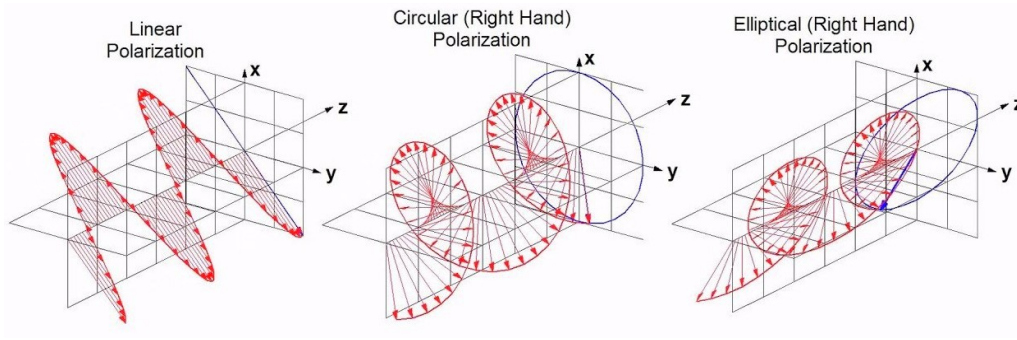
$$\begin{array}{lll} \nabla \cdot \mathbf{E} = 0 & \implies & \mathbf{k} \cdot \mathbf{E}_0 = 0 \\ \nabla \cdot \mathbf{B} = 0 & \implies & \mathbf{k} \cdot \mathbf{B}_0 = 0 \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} & \implies & \mathbf{k} \times \mathbf{E}_0 = \omega \mathbf{B}_0 \\ \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} & \implies & \mathbf{k} \times \mathbf{B}_0 = -\frac{\omega}{c^2} \mathbf{E}_0 \end{array}$$

The fourth equation is redundant because the first and third equations, together with the dispersion relation, imply

$$\mathbf{k} \times \mathbf{B}_0 = \frac{1}{\omega} \mathbf{k} \times (\mathbf{k} \times \mathbf{E}_0) = -\frac{k^2}{\omega} \mathbf{E}_0 = -\frac{\omega}{c^2} \mathbf{E}_0.$$

Suppose $\mathbf{E}_0$ is real, then $\mathbf{B}_0$ is also real and the vectors $\{\mathbf{k}, \mathbf{E}_0, \mathbf{B}_0\}$ form an orthogonal triad. So $\mathbf{E}$ and $\mathbf{B}$ oscillate in fixed directions, which are perpendicular to each other and to the direction of propagation. This is similar to the 1D wave considered previously and corresponds to a linearly polarised wave. Polarisation describes the orientation of the electric field.

Now suppose $\mathbf{E}_0$ is complex and of the form $\mathbf{E}_0 = \mathbf{a} - i\mathbf{b}$ where $\mathbf{a}$ and $\mathbf{b}$ are linearly independent real vectors in the plane perpendicular to $\mathbf{k}$. Then, the direction of $\mathbf{E} = \mathbf{a} \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) + \mathbf{b} \sin(\mathbf{k} \cdot \mathbf{x} - \omega t)$ is not fixed but rotates in the plane perpendicular to $\mathbf{k}$. In general, it traces out an ellipse and the wave is said to be elliptically polarised. The special case of circular polarisation occurs when $|\mathbf{a}| = |\mathbf{b}|$ and $\mathbf{a} \cdot \mathbf{b} = 0$. Then, $\mathbf{E}$ traces out a circle. The polarisation is right-handed if $\mathbf{b} = \hat{\mathbf{k}} \times \mathbf{a}$, and left-handed if $\mathbf{b} = -\hat{\mathbf{k}} \times \mathbf{a}$, where $\hat{\mathbf{k}} = \mathbf{k}/k$.



4.4.3 Reflection from a conductor

Suppose a perfect conductor occupies the half-space $\{x > 0\}$. An incident wave approaching the surface $x = 0$ in the $+x$ direction has the form $\mathbf{E}_{\text{inc}} = \text{Re}(E_0 e^{i(kx - \omega t)} \mathbf{e}_y)$ where we take $k > 0$ and $E_0 \in \mathbb{R}$. Inside the conductor, we have $\mathbf{E} = \mathbf{0}$. Since the tangential components of $\mathbf{E}$ are continuous at the interface $[\mathbf{n} \times \mathbf{E}] = 0$, we require $E_y = 0$ at $x = 0$. This can be satisfied by adding a reflected wave $\mathbf{E}_{\text{ref}} = \text{Re}(-E_0 e^{i(-kx - \omega t)} \mathbf{e}_y)$ propagating in the $-x$ direction. The combined solution is

$$\mathbf{E} = \mathbf{E}_{\text{inc}} + \mathbf{E}_{\text{ref}} = \text{Re}(2iE_0 \sin(kx) e^{-i\omega t} \mathbf{e}_y) = 2E_0 \sin(kx) \sin(\omega t) \mathbf{e}_y$$

which is a standing wave. As an exercise, one can show that $\mathbf{B}$ is discontinuous at $x = 0$ and work out the surface current density $\mathbf{K}(t)$.

4.5 Electromagnetic energy

In electrostatics, we found that the energy per unit volume in $\mathbf{E}$ is $\epsilon_0 |\mathbf{E}|^2 / 2$. In magnetostatics, we found that the energy per unit volume in $\mathbf{B}$ is $|\mathbf{B}|^2 / (2\mu_0)$. We will now derive a general conservation law from Maxwell's equations. Consider the rate of change of the electric and magnetic energy per unit volume.

$$\begin{aligned} \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon_0 |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2 \right) &= \mathbf{E} \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) + \mathbf{B} \cdot \left(\frac{1}{\mu_0} \frac{\partial \mathbf{B}}{\partial t} \right) \\ &= \mathbf{E} \cdot \left(\frac{1}{\mu_0} \nabla \times \mathbf{B} - \mathbf{J} \right) + \mathbf{B} \cdot \left(-\frac{1}{\mu_0} \nabla \times \mathbf{E} \right) \end{aligned}$$

An identity from vector calculus is $\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B})$. Divide by μ_0 and add to the previous equation.

$$\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon_0 |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2 \right) + \nabla \cdot \left(\frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \right) = -\mathbf{E} \cdot \mathbf{J}.$$

This is the energy equation for the electromagnetic field. We can write it as the following.

4.3 – Conservation of energy

$$\frac{\partial w}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J}$$

where $w = \epsilon_0 |\mathbf{E}|^2 / 2 + |\mathbf{B}|^2 / (2\mu_0)$ is the energy density (energy per unit volume) and $\mathbf{S} = (\mathbf{E} \times \mathbf{B}) / \mu_0$ is the energy flux density (energy flux per unit area). $\mathbf{S}$ is known as the Poynting vector.

In the absence of currents, this simplifies to

$$\frac{\partial w}{\partial t} + \nabla \cdot \mathbf{S} = 0$$

and expresses the conservation of electromagnetic energy. Integrate over a time-independent volume V with

surface $\partial V = S$ and use the divergence theorem.

$$\frac{d}{dt} \int_V w dV + \int_S \mathbf{S} \cdot d\mathbf{S} = 0$$

so the rate of change of energy in V is the negative of the flux of energy through S . $\mathbf{E} \cdot \mathbf{J}$ is the rate at which the electromagnetic field loses energy per unit volume by doing work on charged particles via the Lorentz force.

Recall that $\rho = \sum_{i=1}^N q_i \delta(\mathbf{x} - \mathbf{x}_i)$ and $\mathbf{J} = \sum_{i=1}^N q_i \dot{\mathbf{x}}_i \delta(\mathbf{x} - \mathbf{x}_i)$ when we have point charges. The Lorentz force is $\mathbf{F}_i = q_i(\mathbf{E}(\mathbf{x}_i, t) + \dot{\mathbf{x}}_i \times \mathbf{B}(\mathbf{x}_i, t))$ so the rate of work done is

$$\sum_{i=1}^N \mathbf{F}_i \cdot \dot{\mathbf{x}}_i = \sum_{i=1}^N \mathbf{E}(\mathbf{x}_i, t) \cdot (q_i \dot{\mathbf{x}}_i) = \int \mathbf{E} \cdot \mathbf{J} dV.$$

In a conductor, Ohm's law $\mathbf{J} = \sigma \mathbf{E}$ implies $\mathbf{E} \cdot \mathbf{J} = |\mathbf{J}|^2 / \sigma$ which is positive definite and is the rate of Joule heating per unit volume. That is, the rate at which electromagnetic energy is transformed into heat.

Consider an electromagnetic wave with $\mathbf{E} = \text{Re}(\mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)})$ and $\mathbf{B} = \text{Re}(\mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)})$. Using the identity $\text{Re}(A)\text{Re}(B) = \text{Re}(AB + A\bar{B})/2$, we have

$$\begin{aligned} w &= \frac{1}{2} \epsilon_0 |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2 \\ &= \frac{1}{4} \epsilon_0 \text{Re}((\mathbf{E}_0 \cdot \mathbf{E}_0) e^{2i(\mathbf{k} \cdot \mathbf{x} - \omega t)} + |\mathbf{E}_0|^2) + \frac{1}{4\mu_0} \text{Re}((\mathbf{B}_0 \cdot \mathbf{B}_0) e^{2i(\mathbf{k} \cdot \mathbf{x} - \omega t)} + |\mathbf{B}_0|^2) \end{aligned}$$

where $|\mathbf{E}_0|^2 = \mathbf{E}_0 \cdot \bar{\mathbf{E}}_0$. The $e^{2i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ terms average to zero, leaving the average energy density

$$\langle w \rangle = \frac{1}{4} \epsilon_0 |\mathbf{E}_0|^2 + \frac{1}{4\mu_0} |\mathbf{B}_0|^2.$$

Since $\mathbf{B}_0 = (\hat{\mathbf{k}} \times \mathbf{E}_0)/c$ and $\hat{\mathbf{k}} \cdot \mathbf{E}_0 = 0$, the two contributions are equal and we have $\langle w \rangle = \frac{1}{2} \epsilon_0 |\mathbf{E}_0|^2$. Similarly, the Poynting vector is

$$\begin{aligned} \mathbf{S} &= \frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \\ &= \frac{1}{2\mu_0} \text{Re}(\underbrace{(\mathbf{E}_0 \times \mathbf{B}_0) e^{2i(\mathbf{k} \cdot \mathbf{x} - \omega t)}}_{\text{averages to zero}} + \mathbf{E}_0 \times \bar{\mathbf{B}}_0) \\ \langle \mathbf{S} \rangle &= \frac{|\mathbf{E}_0|^2}{2\mu_0 c} \hat{\mathbf{k}} = \frac{1}{2} \epsilon_0 |\mathbf{E}_0|^2 c \hat{\mathbf{k}} = \langle w \rangle c \hat{\mathbf{k}}. \end{aligned}$$

This implies that the electromagnetic wave transports energy at speed c in the direction $\hat{\mathbf{k}}$.

5 Electromagnetism and relativity

5.1 Review of special relativity

5.1.1 Lorentz transformations

In the special theory of relativity, we use spacetime coordinates $X^\mu = (ct, x, y, z) = (ct, \mathbf{x})$ to describe time and position in an inertial frame $\mathcal{S}$. Greek indices such as μ and ν run from 0 to 3 to cover the dimensions of spacetime. Roman indices such as i and j run from 1 to 3 to cover the dimensions of space.

X is the position 4-vector. Under a Lorentz transformation from $\mathcal{S}$ to another inertial frame $\mathcal{S}'$, it transforms

according to $X' = \Lambda X$ where Λ is the 4×4 Lorentz transformation matrix. In index notation,

$$X'^{\mu} = \Lambda^{\mu}_{\nu} X^{\nu}$$

where the summation convention implies a sum from 0 to 3 over the repeated index ν . We have

$$\frac{\partial X'^{\mu}}{\partial X^{\nu}} = \Lambda^{\mu}_{\nu}.$$

An important example of a Lorentz transformation is a Lorentz boost in standard configuration. $\mathcal{S}'$ moves relative to $\mathcal{S}$ with speed v in the x -direction, with $|v| < c$. Then,

$$t' = \gamma \left(t - \frac{vx}{c^2} \right) \quad x' = \gamma(x - vt) \quad y' = y \quad z' = z.$$

The Lorentz factor γ is $\gamma = 1/\sqrt{1 - v^2/c^2}$. The corresponding matrix is

$$\Lambda^{\mu}_{\nu} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

with $\beta = v/c \in (-1, 1)$ and $\gamma = 1/\sqrt{1 - \beta^2} \geq 1$. A more general Lorentz transformation may combine a boost with spatial isometries like rotations and reflections.

5.1.2 Four-vectors

Any 4-vector V transforms as $V' = \Lambda V$ under a Lorentz transformation. The inner product of two 4-vectors V and W is

$$V \cdot W = \eta_{\mu\nu} V^{\mu} W^{\nu}$$

where $\eta_{\mu\nu}$ is the Minkowski metric tensor. It is isotropic, having the same components $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ in any inertial frame. Thus, $V \cdot W = -V^0 W^0 + V^1 W^1 + V^2 W^2 + V^3 W^3$. The inner product is invariant under a Lorentz transformation.

$$V' \cdot W' = \eta'_{\mu\nu} V'^{\mu} W'^{\nu} = \eta_{\mu\nu} \Lambda^{\mu}_{\rho} V^{\rho} \Lambda^{\nu}_{\sigma} W^{\sigma} = \eta_{\rho\sigma} V^{\rho} W^{\sigma} = V \cdot W$$

since the Lorentz transformation matrix satisfies the equation $\eta_{\mu\nu} \Lambda^{\mu}_{\rho} \Lambda^{\nu}_{\sigma} = \eta_{\rho\sigma}$. In matrix notation, $\Lambda^{\top} \eta \Lambda = \eta$ which implies $\det(\Lambda) = \pm 1$.

The square of a 4-vector V is

$$V \cdot V = \eta_{\mu\nu} V^{\mu} V^{\nu} = -V^0 V^0 + V^i V^i.$$

It is also Lorentz-invariant. V is timelike if $V \cdot V < 0$. V is spacelike if $V \cdot V > 0$. V is null or lightlike if $V \cdot V = 0$. For example, the invariant spacetime interval between two events

$$ds^2 = dX \cdot dX = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

5.1.3 Proper time and 4-velocity

The 4-velocity of a massive particle is

$$U = \frac{dX}{d\tau}$$

where τ is the proper time, which is the time expressed by the particle. Since $X^{\mu} = (ct, \mathbf{x})$, we get

$$U^{\mu} = \frac{dt}{d\tau}(c, \mathbf{u}) = \gamma_{\mathbf{u}}(c, \mathbf{u})$$

where $\mathbf{u} = d\mathbf{x}/dt$ is the 3-velocity and $\gamma_{\mathbf{u}} = 1/\sqrt{1 - |\mathbf{u}|^2/c^2}$. Then,

$$U \cdot U = \left[\frac{dt}{d\tau} \right]^2 (-c^2 + |\mathbf{u}|^2) = -c^2$$

is Lorentz-invariant as expected. U is timelike.

The 4-momentum of a massive particle with rest mass $m > 0$ is

$$P^\mu = mU^\mu = m\gamma_{\mathbf{u}}(c, \mathbf{u}) = (E/c, \mathbf{p})$$

where E is the energy and $\mathbf{p}$ is the 3-momentum. Its square is $P \cdot P = -E^2/c^2 + |\mathbf{p}|^2 = -m^2c^2$ consistent with $E = \sqrt{m^2c^4 + |\mathbf{p}|^2c^2}$. In the Newtonian limit $|\mathbf{u}|^2 \ll c^2$, we have $\mathbf{p} = m\gamma_{\mathbf{u}}\mathbf{u} \approx m\mathbf{u}$ and

$$E = m\gamma_{\mathbf{u}}c^2 \approx m \left(1 + \frac{1}{2} \frac{|\mathbf{u}|^2}{c^2} \right) c^2 = mc^2 + \frac{1}{2}m|\mathbf{u}|^2.$$

In the rest frame of the particle, $\mathbf{u} = \mathbf{0}$ and $E = mc^2$, which is the rest energy.

Massless particles such as photons travel at the speed of light in any inertial frame. They have $E = |\mathbf{p}|c$. Their 4-velocity is null.

5.1.4 Covectors and tensors

Let $Y_\mu = \partial f / \partial X^\mu$ be the derivative of a scalar field $f(X)$ on spacetime. How does this transform under a Lorentz transformation? By the chain rule,

$$\begin{aligned} \frac{\partial f}{\partial X'^\mu} &= \frac{\partial X^\nu}{\partial X'^\mu} \frac{\partial f}{\partial X^\nu} \\ Y'_\mu &= (\Lambda^{-1})^\nu{}_\mu Y_\nu \end{aligned}$$

where $(\Lambda^{-1})^\nu{}_\mu = \partial X^\nu / \partial X'^\mu$ is the inverse Lorentz transformation matrix, satisfying

$$\Lambda^\mu{}_\nu (\Lambda^{-1})^\nu{}_\rho = \frac{\partial X'^\mu}{\partial X^\nu} \frac{\partial X^\nu}{\partial X'^\rho} = \frac{\partial X'^\mu}{\partial X'^\rho} = \delta^\mu{}_\rho.$$

The derivative is an example of a covector. Generally, a covector transforms as $Y' = \Lambda^{-1}Y$.

A 4-tensor of type (r, s) is an object whose 4^{r+s} components transform as

$$T'^{\mu_1 \dots \mu_r}{}_{\nu_1 \dots \nu_s} = \Lambda^{\mu_1}{}_{\rho_1} \dots \Lambda^{\mu_r}{}_{\rho_r} (\Lambda^{-1})^{\sigma_1}{}_{\nu_1} \dots (\Lambda^{-1})^{\sigma_s}{}_{\nu_s} T^{\rho_1 \dots \rho_r}{}_{\sigma_1 \dots \sigma_s}$$

under a Lorentz transformation. Any ordering of the up and down indices is possible. Each up index transforms like a vector, and each down index transforms like a covector. A type $(1, 0)$ tensor is a vector. A type $(0, 1)$ tensor is a covector. A type $(0, 0)$ tensor is a scalar, that is a Lorentz invariant. It is straightforward to show the following.

- (i) A linear combination of (r, s) tensors is also an (r, s) tensor. For example, $A^{\mu\nu}{}_{\rho\sigma} = aB^{\mu\nu}{}_{\rho\sigma} + bC^{\mu\nu}{}_{\rho\sigma}$ where a and b are scalars.
- (ii) The outer product of tensors of type (r_1, s_1) and (r_2, s_2) is a tensor of type $(r_1 + r_2, s_1 + s_2)$. For example, $A^{\mu\nu}{}_{\rho\sigma} = B^{\mu\nu}C_{\rho\sigma}$.
- (iii) The derivative of a tensor field of type (r, s) is a tensor field of type $(r, s + 1)$. For example, $A_\mu{}^{\nu\rho}{}_\sigma = \partial_\mu B^{\nu\rho}{}_\sigma$ where ∂_μ means $\partial/\partial X^\mu$.
- (iv) The contraction of an (r, s) tensor with $r, s > 0$ on a pair of its indices produces an $(r - 1, s - 1)$ tensor. For example, $A^\mu{}_\rho = B^{\mu\nu}{}_{\rho\nu}$. This works only if the pair consists of one up and one down index.

The summation convention on spacetime requires the repeated Greek index to appear once up and once down. If we do not follow this, then $\sum_{\mu=0}^3 X^\mu X^\mu = c^2 t^2 + |\mathbf{x}|^2$ would be a scalar, but it is not Lorentz invariant.

A tensor is isotropic if its components are the same in every inertial frame. For example, the metric tensor $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ is an isotropic $(0, 2)$ tensor, and its inverse $\eta^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ is an isotropic $(2, 0)$ tensor. The Kronecker delta symbol $\delta^\mu_\nu = \text{diag}(1, 1, 1, 1)$ is an isotropic $(1, 1)$ tensor. The Levi-Civita epsilon symbol

$$\epsilon_{\mu\nu\rho\sigma} = \begin{cases} +1 & \text{if } (\mu, \nu, \rho, \sigma) \text{ is an even permutation of } (0, 1, 2, 3) \\ -1 & \text{if } (\mu, \nu, \rho, \sigma) \text{ is an odd permutation of } (0, 1, 2, 3) \\ 0 & \text{otherwise} \end{cases}$$

is an isotropic pseudo-tensor of type $(0, 4)$. A pseudo-tensor satisfies the same transformation law as a tensor but with an extra factor of $\det(\Lambda) = \pm 1$.

A tensor equation is one in which every term is a tensor of the same type and transforms identically under a Lorentz transformation. Such an equation is said to be covariant and is valid in any inertial frame.

5.1.5 Vector calculus in spacetime

The spacetime derivative of a scalar field $f(X)$ is a covector field

$$\partial_\mu f = (\partial_0 f, \partial_i f) = \left(\frac{1}{c} \frac{\partial f}{\partial t}, \nabla f \right).$$

The spacetime divergence of a 4-vector field $V(X) = (V^0, \mathbf{v})$ is a scalar field

$$\partial_\mu V^\mu = \partial_0 V^0 + \partial_i V^i = \frac{1}{c} \frac{\partial V^0}{\partial t} + \nabla \cdot \mathbf{v}.$$

In $\mathbb{R}^3$, the curl involves the antisymmetric part of the derivatives of a vector field. The spacetime equivalent is the antisymmetric derivative of the covector field $Y(X)$, a tensor field of type $(0, 2)$

$$\partial_\mu Y_\nu - \partial_\nu Y_\mu.$$

The generalisation of the Laplacian ∇^2 is the d'Alembertian, a similar differential operator

$$\square = \eta^{\mu\nu} \partial_\mu \partial_\nu = -\partial_0 \partial_0 + \partial_i \partial_i = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2.$$

The wave equation for a scalar field $f(X)$ is $\square f = 0$. This is a scalar equation only for waves that travel at c .

5.1.6 Raising and lowering

The metric tensor provides an isomorphism between vectors (V^μ) and covectors (V_μ) .

$$V^\mu = \eta^{\mu\nu} V_\nu \qquad V_\mu = \eta_{\mu\nu} V^\nu$$

These operations are known as raising and lowering indices. Raising or lowering simply changes the sign of the time component, while leaving the spatial components unchanged. So, raising or lowering a Roman index changes nothing. For example, lowering the 4-velocity $U^\mu = \gamma_{\mathbf{u}}(c, \mathbf{u})$ gives $U_\mu = \gamma_{\mathbf{u}}(-c, \mathbf{u})$. Or the derivative $\partial_\mu f = ((1/c)\partial f/\partial t, \nabla f)$ and $\partial^\mu f = (-(1/c)\partial f/\partial t, \nabla f)$.

The inner product of two 4-vectors can be written variously as

$$V \cdot W = \eta_{\mu\nu} V^\mu W^\nu = V_\nu W^\nu = V^\mu W_\mu = \eta^{\mu\nu} V_\mu W_\nu.$$

Raising and lowering can be applied to any tensor index. For example, $T_\mu{}^\nu = \eta_{\mu\rho} T^{\rho\nu}$ or $T^\mu{}_\nu = \eta_{\nu\rho} T^{\mu\rho}$. We can show that $\eta^\mu{}_\nu = \delta^\mu_\nu$. A tensor may be symmetric or antisymmetric on a pair of its indices if they are both up

or both down. We can show that this property is preserved if both indices are raised or lowered.

5.2 Charge conservation

The equation of charge conservation $\partial\rho/\partial t + \nabla \cdot \mathbf{J} = 0$ can be written in the form of a spacetime divergence

5.1 – Charge conservation

$$\partial_\mu J^\mu = 0$$

where $J^\mu = (\rho c, \mathbf{J})$ is the charge-current density 4-vector.

This is a scalar and Lorentz invariant equation, valid in any inertial frame.

Consider a Lorentz boost in standard configuration. Suppose in frame S , we have a uniform distribution of charged particles at rest, with charge q and number density n . Then, $\rho = nq$ and $\mathbf{J} = \mathbf{0}$, giving $J^\mu = (nqc, \mathbf{0})$ in S . In frame S' , the particles have 3-velocity $(-v, 0, 0)$. Their number density increases to γn because of length contraction in the x -direction. So, $\rho' = \gamma nq$ and $\mathbf{J}' = \gamma nq(-v, 0, 0)$, giving $J'^\mu = (\gamma nqc, -\gamma nqv, 0, 0)$ in S' . These are consistent with the Lorentz transformation of a 4-vector.

Now suppose that in frame S , we have a wire in which particles of charge $+q$ and number density n_+ move along the wire with 3-velocity $(u, 0, 0)$, and particles of charge $-q$ and number density $n_- = n_+$ move with 3-velocity $(-u, 0, 0)$. The net charge density is $\rho = n_+q - n_-q = 0$ and the net current density is

$$\mathbf{J} = n_+q(u, 0, 0) - n_-q(-u, 0, 0) = (2n_+qu, 0, 0).$$

The wire is uncharged but carries a current. So, $\mathbf{E} = \mathbf{0}$ and $\mathbf{B} \neq \mathbf{0}$. In S' , the $+$ and $-$ particles have different x -velocities

$$\frac{\pm u - v}{1 \mp uv/c^2}$$

and therefore different charge densities because of length contraction. The wire has a net charge and therefore $\mathbf{E}' \neq \mathbf{0}$. So a magnetic field in one frame can produce an electric field in another, and vice versa.

5.3 The electromagnetic tensor

The relativistic equation of motion for a massive particle is

$$\frac{dP^\mu}{d\tau} = f^\mu$$

where τ is the proper time and f^μ is the 4-force. Since $P \cdot P = -m^2c^2$ is constant, f^μ must satisfy $f \cdot P = 0$. In detail,

$$\frac{d}{d\tau}(\eta_{\mu\nu}P^\mu P^\nu) = \eta_{\mu\nu} \left(\frac{dP^\mu}{d\tau} P^\nu + P^\mu \frac{dP^\nu}{d\tau} \right) = \eta_{\mu\nu} (f^\mu P^\nu + P^\mu f^\nu) = 2\eta_{\mu\nu} f^\mu P^\nu = 2f_\mu P^\mu = 0.$$

Now can we express the Lorentz force as a 4-vector? It should be proportional to q and linear in U^μ . Assume $f_\mu = qF_{\mu\nu}U^\nu$ for some $(0,2)$ tensor $F_{\mu\nu}$. Then, $f \cdot P = 0$ implies $qF_{\mu\nu}U^\mu U^\nu = 0$ and is satisfied for all U if $F_{\mu\nu}$ is antisymmetric. The spatial components of f_μ are $f_i = q(F_{i0}U^0 + F_{ij}U^j) = \gamma_{\mathbf{u}}q(F_{i0}c + F_{ij}u_j)$. For this to agree with $\gamma_{\mathbf{u}}q(E_i + \epsilon_{ijk}u_j B_k)$, where the factor of $\gamma_{\mathbf{u}}$ comes from converting d/dt into $d/d\tau$, we require $F_{i0} = E_i/c$ and $F_{ij} = \epsilon_{ijk}B_k$. By antisymmetry, $F_{00} = 0$ and $F_{0i} = -E_i/c$. Thus,

5.2 – Electromagnetic field tensor

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & B_z & -B_y \\ E_y/c & -B_z & 0 & B_x \\ E_z/c & B_y & -B_x & 0 \end{pmatrix}$$

Neither $\mathbf{E}$ nor $\mathbf{B}$ is the spatial part of a 4-vector. However, for a particle with 4-velocity U , $F_{\mu\nu}U^\nu$ is a covector that equals $(0, \mathbf{E})$ in the particle's rest frame with $U = (c, \mathbf{0})$.

By construction, the spatial components of $dP^\mu/d\tau = f^\mu$ give the relativistic equation of motion

$$\frac{d\mathbf{p}}{d\tau} = \gamma_{\mathbf{u}} q (\mathbf{E} + \mathbf{u} \times \mathbf{B})$$

where $\mathbf{p} = \gamma_{\mathbf{u}} m \mathbf{u}$ is the relativistic 3-momentum. Equivalently,

$$\frac{d\mathbf{p}}{dt} = q (\mathbf{E} + \mathbf{u} \times \mathbf{B}) = \mathbf{F}$$

where $\mathbf{F}$ is the 3-force. The time component is

$$\frac{dP^0}{d\tau} = f^0 = -f_0 = -q F_{0i} U^i = q \frac{E_i}{c} \gamma_{\mathbf{u}} u_i$$

which is equivalent to

$$\frac{dE}{d\tau} = q \gamma_{\mathbf{u}} \mathbf{E} \cdot \mathbf{u} \implies \frac{dE}{dt} = q \mathbf{E} \cdot \mathbf{u} = \mathbf{F} \cdot \mathbf{u}$$

where E is the energy.

5.4 The 4-potential

We previously wrote $\mathbf{E} = -\nabla\Phi - \partial\mathbf{A}/\partial t$ and $\mathbf{B} = \nabla \times \mathbf{A}$ in terms of scalar and vector potentials. What does this mean for $F_{\mu\nu}$?

$$F_{i0} = \frac{E_i}{c} = -\frac{1}{c} \frac{\partial\Phi}{\partial x_i} - \frac{1}{c} \frac{\partial A_i}{\partial t} = \partial_i(-\Phi/c) - \partial_0 A_i$$

and

$$F_{ij} = \epsilon_{ijk} B_k = \epsilon_{ijk} \epsilon_{klm} \partial_l A_m = \partial_i A_j - \partial_j A_i.$$

We interpret $F_{\mu\nu}$ as the antisymmetric derivative $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ of the covector $A_\mu = (-\Phi/c, \mathbf{A})$. The related 4-vector is the 4-potential

$$A^\mu = \left(\frac{\Phi}{c}, \mathbf{A} \right).$$

A gauge transformation $\tilde{\mathbf{A}} = \mathbf{A} + \nabla\chi$ and $\tilde{\Phi} = \Phi - \partial\chi/\partial t$ corresponds to replacing A_μ with $\tilde{A}_\mu = A_\mu + \partial_\mu\chi$. That is, adding $\partial_\mu\chi = ((1/c)\partial\chi/\partial t, \nabla\chi)$ to $A_\mu = (-\Phi/c, \mathbf{A})$. This leaves $F_{\mu\nu}$ unaffected because $\partial_\mu(\partial_\nu\chi) - \partial_\nu(\partial_\mu\chi) = 0$. The spacetime equivalent of the Coulomb gauge is the Lorentz gauge in which

$$\partial_\mu A^\mu = 0.$$

If $\partial_\mu A^\mu \neq 0$, then we can make a gauge transformation such that $\partial_\mu \tilde{A}^\mu = 0$ by choosing χ to be the solution of the forced wave equation $-\square\chi = \partial_\mu A^\mu$. Note that $\partial_\mu \partial^\mu \chi = \eta^{\mu\nu} \partial_\mu \partial_\nu \chi = \square\chi$.

5.5 Lorentz transformation of $\mathbf{E}$ and $\mathbf{B}$

Since $F_{\mu\nu}$ is a $(0,2)$ tensor, it transform as $F'_{\mu\nu} = (\Lambda^{-1})^\rho{}_\mu (\Lambda^{-1})^\sigma{}_\nu F_{\rho\sigma}$. In matrix notation, $F' = (\Lambda^{-1})^\top F \Lambda^{-1}$. For a Lorentz boost in standard configuration,

$$\Lambda^{-1} = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

We can check that

$$\begin{aligned} F'_{01} &= F_{01} & F'_{02} &= \gamma(F_{02} + \beta F_{12}) & F'_{03} &= \gamma(F_{03} + \beta F_{13}) \\ F'_{12} &= \gamma(F_{12} + \beta F_{02}) & F'_{13} &= \gamma(F_{13} + \beta F_{03}) & F'_{23} &= F_{23}. \end{aligned}$$

We can then deduce

$$\begin{aligned} E'_x &= E_x & E'_y &= \gamma(E_y - vB_z) & E'_z &= \gamma(E_z + vB_y) \\ B'_x &= B_x & B'_y &= \gamma\left(B_y + \frac{v}{c^2}E_z\right) & B'_z &= \gamma\left(B_z - \frac{v}{c^2}E_y\right) \end{aligned}$$

More generally, if $\mathbf{v}$ is the velocity of $\mathcal{S}'$ with respect to $\mathcal{S}$, then

$$\begin{aligned} \mathbf{E}'_{\parallel} &= \mathbf{E}_{\parallel} & \mathbf{E}'_{\perp} &= \gamma(\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}_{\perp}) \\ \mathbf{B}'_{\parallel} &= \mathbf{B}_{\parallel} & \mathbf{B}'_{\perp} &= \gamma\left(\mathbf{B}_{\perp} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}_{\perp}\right) \end{aligned}$$

where we decompose $\mathbf{E}$ and $\mathbf{B}$ into components parallel and perpendicular to $\mathbf{v}$, so $\mathbf{E} = \mathbf{E}_{\parallel} + \mathbf{E}_{\perp}$ and $\mathbf{B} = \mathbf{B}_{\parallel} + \mathbf{B}_{\perp}$. $F_{\mu\nu}$ can be doubly raised to make

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix} = -F^{\nu\mu}.$$

Thus, $F^{00} = 0$, $F^{0i} = E_i/c$, $F^{i0} = -E_i/c$, and $F^{ij} = \epsilon_{ijk}B_k$. We can find two Lorentz-invariant combinations of $\mathbf{E}$ and $\mathbf{B}$ by forming scalars from $F_{\mu\nu}$.

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= -2(|\mathbf{E}|^2 - c^2|\mathbf{B}|^2)/c^2 \\ \epsilon_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma} &= 8(\mathbf{E} \cdot \mathbf{B})/c \end{aligned}$$

Thus, $|\mathbf{E}|^2 - c^2|\mathbf{B}|^2$ is a scalar and $\mathbf{E} \cdot \mathbf{B}$ is a pseudoscalar. For example, if $\mathbf{E} \perp \mathbf{B}$ in one inertial frame, then they are perpendicular in any frame.

5.6 Maxwell's equations

How do we write Maxwell's equations in a covariant form? Using $F^{0i} = -F^{i0} = E_i/c$, $F^{ij} = \epsilon_{ijk}B_k$, $J^0 = \rho c$, $J^i = J_i$, and $c^2 = 1/(\mu_0\epsilon_0)$, (M1) becomes

$$c\partial_i F^{0i} = \frac{J^0}{\epsilon_0 c} \implies \partial_i F^{0i} = \mu_0 J^0$$

and (M4) becomes

$$\epsilon_{ijk}\partial_j B_k = \mu_0 J_i + \frac{1}{c}\partial_0 E_i \implies \partial_0 F^{i0} + \partial_j F^{ij} = \mu_0 J^i.$$

These are the components of the 4-vector equation

$$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu.$$

Now $F_{ij} = \epsilon_{ijk} B_k$ so $B_i = \epsilon_{ijk} F_{jk}/2$. So, (M2) becomes

$$\partial_i B_i = \frac{1}{2} \epsilon_{ijk} \partial_i F_{jk} = \partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = 0 \implies \partial_i F_{jk} + \partial_j F_{ki} + \partial_k F_{ij} = 0.$$

This is trivially satisfied if any of the i, j, k coincide since F is antisymmetric. (M3) becomes

$$\epsilon_{ijk} \partial_j E_k = -c \partial_0 B_i \implies c \epsilon_{ijk} \partial_j F_{k0} + \frac{c}{2} \epsilon_{ijk} \partial_0 F_{jk} = 0.$$

Multiply by ϵ_{ilm} and divide by c gives $\partial_l F_{m0} - \partial_m F_{l0} + \partial_0 F_{lm} = 0$. Using antisymmetry and relabelling, $\partial_i F_{j0} + \partial_j F_{0i} + \partial_0 F_{ij} = 0$. Combine them into the covariant form

$$\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0.$$

Alternatively, we can write this as a 4-vector equation

$$\epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0.$$

Its 0-component is $\epsilon^{0ijk} \partial_i F_{jk} = -\epsilon_{ijk} \partial_i F_{jk} = 0$ which is equivalent to $\partial_i B_i = 0$. Its i -component is

$$\epsilon^{i0jk} \partial_0 F_{jk} + \epsilon^{ij0k} \partial_j F_{0k} + \epsilon^{ijk0} \partial_j F_{k0} = \epsilon_{ijk} \partial_0 F_{jk} - \epsilon_{ijk} \partial_j F_{0k} + \epsilon_{ijk} \partial_j F_{k0} = \epsilon_{ijk} \partial_0 F_{jk} + 2\epsilon_{ijk} \partial_j F_{k0} = 0$$

which is equivalent to (M3) as above.

In summary, we have

5.3 – Maxwell's equations in covariant form

$$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu \quad (C1)$$

$$\partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} = 0 \quad \text{or} \quad \epsilon^{\mu\nu\rho\sigma} \partial_\nu F_{\rho\sigma} = 0 \quad (C2)$$

The equation of motion of a charged particle is

$$\frac{dP^\mu}{d\tau} = f^\mu$$

with $P^\mu = mU^\mu$ and $f^\mu = qF_{\mu\nu}U^\nu$. Charge conservation follows easily from (C1),

$$\frac{1}{\mu_0} \partial_\mu \partial_\nu F^{\mu\nu} = \partial_\mu J^\mu = 0$$

since $\partial_\mu \partial_\nu$ is symmetric while $F^{\mu\nu}$ is antisymmetric.

If we write $F_{\mu\nu}$ in terms of the 4-potential as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, then (C2) is automatically satisfied since $\partial_\mu \partial_\nu A_\rho - \partial_\nu \partial_\mu A_\rho + \partial_\nu \partial_\rho A_\mu - \partial_\rho \partial_\nu A_\mu + \partial_\rho \partial_\mu A_\nu - \partial_\mu \partial_\rho A_\nu = 0$ and $\epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\rho A_\sigma - \epsilon^{\mu\nu\rho\sigma} \partial_\nu \partial_\sigma A_\rho = 0$, while (C1) becomes

$$\partial_\nu (\partial^\mu A^\nu) - \partial_\nu (\partial^\nu A^\mu) = \mu_0 J^\mu.$$

In the Lorentz gauge, the first term vanishes since $\partial_\nu (\partial^\mu A^\nu) = \partial^\mu (\partial_\nu A^\nu) = 0$, leaving

$$-\square A^\mu = \mu_0 J^\mu.$$

This is an inhomogeneous wave equation for the 4-potential, with a source proportional to the charge-current density. In the absence of charges and currents, the equation describes free electromagnetic waves. In a time-independent situation, it reduces to $-\nabla^2 \Phi = \rho/\epsilon_0$ and $-\nabla^2 \mathbf{A} = \mu_0 \mathbf{J}$ as in electrostatic and magnetostatics.

More generally, it describes how electromagnetic waves are generated by time-dependent charge and current distributions.